

JUNO-X MIDI Implementation

Model: JUNO-X

Date: April 26, 2022

Version: 1.10

1. Data Reception

■ Channel Voice Messages

● Note Off

Status	2nd byte	3rd byte
8nH	kkH	vvH
9nH	kkH	00H

n = MIDI channel number: 0H - FH (ch.1 - ch.16)

kk = Note number: 00H - 7FH (0 - 127)

vv = Note off velocity: 00H - 7FH (0 - 127)

● Note On

Status	2nd byte	3rd byte
9nH	kkH	vvH

n = MIDI channel number: 0H - FH (ch.1 - ch.16)

kk = Note number: 00H - 7FH (0 - 127)

vv = Note on velocity: 01H - 7FH (1 - 127)

● Polyphonic Key Pressure

Status	2nd byte	3rd byte
AnH	kkH	vvH

n = MIDI channel number: 0H - FH (ch.1 - ch.16)

kk = note number: 00H - 7FH (0 - 127)

vv = Polyphonic Key Pressure: 00H - 7FH (0 - 127)

* Not received when the Rx Poly Pres parameter (SYSTEM: MIDI: MIDI RX) is OFF.

● Control Change

* If you select a controller number that corresponds to an MFX Src 1-4 parameter, the respective effect is applied according to the setting.

○ Bank Select (Controller number 0, 32)

Status	2nd byte	3rd byte
BnH	00H	mmH
BnH	20H	11H

n = MIDI channel number: 0H - FH (ch.1 - ch.16)

mm, ll = Bank number: 00 00H - 7F 7FH (bank.1 - bank.16384)

* Not received when the Rx Bank parameter (SYSTEM: MIDI RX) is OFF.

* Not received when the Rx Bank parameter (SCENE PART EDIT: MIDI) is OFF.

The Scenes corresponding to each Bank Select are as follows.

BANK SELECT	PROGRAM	GROUP	NUMBER
MSB LSB	NUMBER		
085 000 - 001	001 - 128	SCENE	1 - 256

The Tones corresponding to each Bank Select are as follows.

BANK SELECT	PROGRAM	GROUP	NUMBER
MSB LSB	NUMBER		
071 003 - 004	001 - 128	User Tone	0001 - 0256
071 067	001 - 035	PR-X	0001 - 0035
069	001 - 005	RD-PIANO	0001 - 0005
070	001 - 002	VOCODER	0001 - 0002
087 064	001 - 128	PR-B	0001 - 0128
:	:	:	:
067	001 - 075		0385 - 0459
068	001 - 128	PR-C	0001 - 0128
069	001 - 128	PR-D	0001 - 0128
:	:	:	:
077	001 - 085		1025 - 1109
078	001 - 128	XV-5080	0001 - 0128
:	:	:	:
084	001 - 128		0769 - 0896
085	001 - 128	COMMON	0001 - 0128
:	:	:	:
091	001 - 069		0768 - 0837
092	001 - 128	PR-A	0001 - 0128
093	001 - 111		0129 - 0239
097 066	001 - 122	JUNO-106	0001 - 0122
074	001 - 128	JUNO-60	0001 - 0128
075	001 - 009		0129 - 0137
076	001 - 128	JUNO-X	0001 - 0128
077	001 - 017		0129 - 0145
086 064	001 - 017	PR-A DRUM	0001 - 0017
065	001 - 074	CMN DRUM	0001 - 0074

○ Modulation (Controller number 1)

Status	2nd byte	3rd byte
BnH	01H	vvH

n = MIDI channel number: 0H - FH (ch.1 - ch.16)

vv = Modulation depth: 00H - 7FH (0 - 127)

* Not received when the Rx Mod parameter (SCENE PART EDIT: MIDI) is OFF.

○ Breath type (Controller number 2)

Status	2nd byte	3rd byte
BnH	02H	vvH

n = MIDI channel number: 0H - FH (ch.1 - ch.16)

vv = Control value: 00H - 7FH (0 - 127)

○ Foot Type (Controller number 4)

Status	2nd byte	3rd byte
BnH	04H	vvH

n = MIDI channel number: 0H - FH (ch.1 - ch.16)

vv = Control value: 00H - 7FH (0 - 127)

○ Portamento Time (Controller number 5)

Status	2nd byte	3rd byte
BnH	05H	vvH

n = MIDI channel number: 0H - FH (ch.1 - ch.16)

vv = Portamento Time: 00H - 7FH (0 - 127)

* The Porta Time parameter (SCENE PART EDIT: CTRL) will change.

○ Data Entry (Controller number 6, 38)

Status	2nd byte	3rd byte
BnH	06H	mmH
BnH	26H	11H

n = MIDI channel number: 0H - FH (ch.1 - ch.16)

mm, ll = the value of the parameter specified by RPN/NRPN

mm = MSB, ll = LSB

o Volume (Controller number 7)

Status	2nd byte	3rd byte
BnH	07H	vvH

n = MIDI channel number: 0H - FH (ch.1 - ch.16)

vv = Volume: 00H - 7FH (0 - 127)

- * Not received when the Rx Volume parameter (SCENE PART EDIT: MIDI) is OFF.
- * The Part Level parameter (SCENE PART EDIT) will change.

o Panpot (Controller number 10)

Status	2nd byte	3rd byte
BnH	0AH	vvH

n = MIDI channel number: 0H - FH (ch.1 - ch.16)

vv = Panpot: 00H - 40H - 7FH (Left - Center - Right)

- * Not received when the Rx Pan parameter (SCENE PART EDIT: MIDI) is OFF.
- * The Pan parameter (SCENE PART EDIT) will change.

o Expression (Controller number 11)

Status	2nd byte	3rd byte
BnH	0BH	vvH

n = MIDI channel number: 0H - FH (ch.1 - ch.16)

vv = Expression: 00H - 7FH (0 - 127)

- * Not received when the Rx Exp parameter (SCENE PART EDIT: MIDI) is OFF.

o General Purpose Controller 1 (Controller number 16)

Status	2nd byte	3rd byte
BnH	10H	vvH

n = MIDI channel number: 0H - FH (ch.1 - ch.16)

vv = Control value: 00H - 7FH (0 - 127)

o General Purpose Controller 2 (Controller number 17)

Status	2nd byte	3rd byte
BnH	11H	vvH

n = MIDI channel number: 0H - FH (ch.1 - ch.16)

vv = Control value: 00H - 7FH (0 - 127)

o General Purpose Controller 3 (Controller number 18)

Status	2nd byte	3rd byte
BnH	12H	vvH

n = MIDI channel number: 0H - FH (ch.1 - ch.16)

vv = Control value: 00H - 7FH (0 - 127)

o General Purpose Controller 4 (Controller number 19)

Status	2nd byte	3rd byte
BnH	13H	vvH

n = MIDI channel number: 0H - FH (ch.1 - ch.16)

vv = Control value: 00H - 7FH (0 - 127)

o Hold 1 (Controller number 64)

Status	2nd byte	3rd byte
BnH	40H	vvH

n = MIDI channel number: 0H - FH (ch.1 - ch.16)

vv = Control value: 00H - 7FH (0 - 127)

0-63 = OFF, 64-127 = ON

- * Not received when the Rx Hold-1 parameter (SCENE PART EDIT: MIDI) is OFF.

o Portamento (Controller number 65)

Status	2nd byte	3rd byte
BnH	41H	vvH

n = MIDI channel number: 0H - FH (ch.1 - ch.16)

vv = Control value: 00H - 7FH (0 - 127)

0-63 = OFF, 64-127 = ON

- * The Porta Sw parameter (SCENE PART EDIT: CTRL) will change.

o Sostenuto (Controller number 66)

Status	2nd byte	3rd byte
BnH	42H	vvH

n = MIDI channel number: 0H - FH (ch.1 - ch.16)

vv = Control value: 00H - 7FH (0 - 127)

0-63 = OFF, 64-127 = ON

o Soft (Controller number 67)

Status	2nd byte	3rd byte
BnH	43H	vvH

n = MIDI channel number: 0H - FH (ch.1 - ch.16)

vv = Control value: 00H - 7FH (0 - 127)

o Legato Foot Switch (Controller number 68)

Status	2nd byte	3rd byte
BnH	44H	vvH

n = MIDI channel number: 0H - FH (ch.1 - ch.16)

vv = Control value: 00H - 7FH (0 - 127)

0-63 = OFF, 64-127 = ON

- * The Legato Sw parameter (SCENE PART EDIT: CTRL) will change.

o Resonance (Controller number 71)

Status	2nd byte	3rd byte
BnH	47H	vvH

n = MIDI channel number: 0H - FH (ch.1 - 16)

vv = Resonance value (relative change): 00H - 40H - 7FH (-64 - 0 - +63)

- * The Resonance parameter (SCENE PART EDIT: MODIFY) will change.

o Release Time (Controller number 72)

Status	2nd byte	3rd byte
BnH	48H	vvH

n = MIDI channel number: 0H - FH (ch.1 - 16)

vv = Release Time value (relative change): 00H - 40H - 7FH (-64 - 0 - +63)

- * The Release parameter (SCENE PART EDIT: MODIFY) will change.

o Attack time (Controller number 73)

Status	2nd byte	3rd byte
BnH	49H	vvH

n = MIDI channel number: 0H - FH (ch.1 - 16)

vv = Attack time value (relative change): 00H - 40H - 7FH (-64 - 0 - +63)

* The Attack parameter (SCENE PART EDIT: MODIFY) will change.

o Cutoff (Controller number 74)

Status	2nd byte	3rd byte
BnH	4AH	vvH

n = MIDI channel number: 0H - FH (ch.1 - 16)

vv = Cutoff value (relative change): 00H - 40H - 7FH (-64 - 0 - +63)

* The Cutoff parameter (SCENE PART EDIT: MODIFY) will change.

o Decay Time (Controller number 75)

Status	2nd byte	3rd byte
BnH	4BH	vvH

n = MIDI channel number: 0H - FH (ch.1 - 16)

vv = Decay Time value (relative change): 00H - 40H - 7FH (-64 - 0 - +63)

* The Decay parameter (SCENE PART EDIT: MODIFY) will change.

o Vibrato Rate (Controller number 76)

Status	2nd byte	3rd byte
BnH	4CH	vvH

n = MIDI channel number: 0H - FH (ch.1 - 16)

vv = Vibrato Rate value (relative change): 00H - 40H - 7FH (-64 - 0 - +63)

* The Vib Rate parameter (SCENE PART EDIT: MODIFY) will change.

o Vibrato Depth (Controller number 77)

Status	2nd byte	3rd byte
BnH	4DH	vvH

n = MIDI channel number: 0H - FH (ch.1 - 16)

vv = Vibrato Depth Value (relative change): 00H - 40H - 7FH (-64 - 0 - +63)

* The Vib Depth parameter (SCENE PART EDIT: MODIFY) will change.

o Vibrato Delay (Controller number 78)

Status	2nd byte	3rd byte
BnH	4EH	vvH

n = MIDI channel number: 0H - FH (ch.1 - 16)

vv = Vibrato Delay value (relative change): 00H - 40H - 7FH (-64 - 0 - +63)

* The Vib Delay parameter (SCENE PART EDIT: MODIFY) will change.

o General Purpose Controller 5 (Controller number 80)

Status	2nd byte	3rd byte
BnH	50H	vvH

n = MIDI channel number: 0H - FH (ch.1 - ch.16)

vv = Control value: 00H - 7FH (0 - 127)

o General Purpose Controller 6 (Controller number 81)

Status	2nd byte	3rd byte
BnH	51H	vvH

n = MIDI channel number: 0H - FH (ch.1 - ch.16)

vv = Control value: 00H - 7FH (0 - 127)

o General Purpose Controller 7 (Controller number 82)

Status	2nd byte	3rd byte
BnH	52H	vvH

n = MIDI channel number: 0H - FH (ch.1 - ch.16)

vv = Control value: 00H - 7FH (0 - 127)

o General Purpose Controller 8 (Controller number 83)

Status	2nd byte	3rd byte
BnH	53H	vvH

n = MIDI channel number: 0H - FH (ch.1 - ch.16)

vv = Control value: 00H - 7FH (0 - 127)

o Portamento control (Controller number 84)

Status	2nd byte	3rd byte
BnH	54H	kkH

n = MIDI channel number: 0H - FH (ch.1 - ch.16)

kk = source note number: 00H - 7FH (0 - 127)

* A Note-on received immediately after a Portamento Control message will change continuously in pitch, starting from the pitch of the Source Note Number.

* If a voice is already sounding for a note number identical to the Source Note Number, this voice will continue sounding (i.e., legato) and will, when the next Note-on is received, smoothly change to the pitch of that Note-on.

* The rate of the pitch change caused by Portamento Control is determined by the Portamento Time value.

o General Purpose Effect 1 (Reverb Send Level)

(Controller number 91)

Status	2nd byte	3rd byte
BnH	5BH	vvH

n = MIDI channel number: 0H - FH (ch.1 - ch.16)

vv = Reverb Send Level: 00H - 7FH (0 - 127)

* The Rev Send parameter (SCENE PART EDIT: MODIFY) will change.

o General Purpose Effect 3 (Chorus Send Level)

(Controller number 93)

Status	2nd byte	3rd byte
BnH	5DH	vvH

n = MIDI channel number: 0H - FH (ch.1 - ch.16)

vv = Chorus Send Level: 00H - 7FH (0 - 127)

* The Cho Send parameter (SCENE PART EDIT: MODIFY) will change.

o RPN MSB/LSB (Controller number 100, 101)

Status	2nd byte	3rd byte
BnH	65H	mmH
BnH	64H	11H

n = MIDI channel number: 0H - FH (ch.1 - ch.16)

mm = upper byte (MSB) of parameter number specified by RPN

11 = lower byte (LSB) of parameter number specified by RPN

<<< RPN >>>

Control Changes include RPN (Registered Parameter Numbers), which are extended.

When using RPNs, first RPN (Controller numbers 100 and 101; they can be sent in any order) should be sent in order to select the parameter, then Data Entry (Controller numbers 6 and 38) should be sent to set the value. Once RPN messages are received, Data Entry messages that is received at the same MIDI channel after that are recognized as changing toward the value of the RPN messages. In order not to make any mistakes, transmitting RPN Null is recommended after setting parameters you need.

This device receives the following RPNs.

RPN	Data entry	Notes
MSB, LSB	MSB, LSB	
00H, 00H	mmH, 11H	Pitch Bend Sensitivity mm: 00H - 18H (0 - 24 semitones) 11: ignored (processed as 00H) Up to 2 octave can be specified in semitone steps.

* The Pitch Bend Range (Bend Range) parameter (PART VIEW: Pitch) will change.

00H, 01H	mmH, 11H	Channel Fine Tuning mm, 11: 20 00H - 40 00H - 60 00H (-4096 x 100 / 8192 - 0 - +4096 x 100 / 8192 cent)
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* The Fine Tune parameter (SCENE PART EDIT: PITCH) will change.

00H, 02H	mmH, 11H	Channel Coarse Tuning mm: 10H - 40H - 70H (-48 - 0 - +48 semitones) 11: ignored (processed as 00H)
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* The Coarse Tune parameter (SCENE PART EDIT: PITCH) will change.

7FH, 7FH	---, ---	RPN null RPN and NRPN will be set as "unspecified." Once this setting has been made, subsequent Parameter values that were previously set will not change. mm, 11: ignored
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● Program Change

Status	2nd byte
CnH	ppH

n = MIDI channel number: 0H - FH (ch.1 - ch.16)

pp = Program number: 00H - 7FH (prog.1 - prog.128)

* Not received when the Rx PC parameter (SYSTEM: MIDI RX) is OFF.

* Not received when the Rx PC parameter (SCENE PART EDIT: MIDI) is OFF.

● Channel Pressure

Status	2nd byte
DnH	vvH

n = MIDI channel number: 0H - FH (ch.1 - ch.16)

vv = Channel Pressure: 00H - 7FH (0 - 127)

* Not received when the Rx Ch Pres parameter (SCENE PART EDIT: MIDI) is OFF.

● Pitch Bend Change

Status	2nd byte	3rd byte
EnH	11H	mmH

n = MIDI channel number: 0H - FH (ch.1 - ch.16)

mm, 11 = Pitch Bend value: 00 00H - 40 00H - 7F 7FH (-8192 - 0 - +8191)

* Not received when the Rx Bend parameter (SCENE PART EDIT: MIDI) is OFF.

■ Channel Mode Messages

● All Sounds Off

(Controller number 120)

Status	2nd byte	3rd byte
BnH	78H	00H

n = MIDI channel number: 0H - FH (ch.1 - ch.16)

* When this message is received, all notes currently sounding on the corresponding channel will be turned off.

● Reset All Controllers

(Controller number 121)

(Controller number 121)

Status	2nd byte	3rd byte
BnH	79H	00H

n = MIDI channel number: 0H - FH (ch.1 - ch.16)

* When this message is received, the following controllers will be set to their reset values.

Controller	Reset value
Pitch Bend Change	+/-0 (center)
Polyphonic Key Pressure	0 (off)
Channel Pressure	0 (off)
Modulation	0 (off)
Breath Type	0 (min)
Foot Type	0 (min)
Expression	127 (max) However the controller will be at minimum.
Hold 1	0 (off)
Sostenuto	0 (off)
Soft	0 (off)
RPN	unset; previously set data will not change
NRPN	unset; previously set data will not change

● All Notes Off

(Controller number 123)

Status	2nd byte	3rd byte
BnH	7BH	00H

n = MIDI channel number: 0H - FH (ch.1 - ch.16)

* When All Notes Off is received, all notes on the corresponding channel will be turned off. However, if Hold 1 or Sostenuto is ON, the sound will be continued until these are turned off.

● OMNI OFF (Controller number 124)

Status	2nd byte	3rd byte
BnH	7CH	00H

n = MIDI channel number: 0H - FH (ch.1 - ch.16)

* The same processing will be carried out as when All Notes Off is received.

● OMNI ON (Controller number 125)

Status	2nd byte	3rd byte
BnH	7DH	00H

n = MIDI channel number: 0H - FH (ch.1 - ch.16)

* The same processing will be carried out as when All Notes Off is received. OMNI ON will not be turned on.

● MONO (Controller number 126)

Status	2nd byte	3rd byte
BnH	7EH	mmH

n = MIDI channel number: 0H - FH (ch.1 - ch.16)

mm = mono number: 00H - 10H (0 - 16)

* The same processing will be carried out as when All Notes Off is received.
* The Mono/Poly parameter (SCENE PART EDIT: CTRL) will change.

● POLY (Controller number 127)

Status	2nd byte	3rd byte
BnH	7FH	00H

n = MIDI channel number: 0H - FH (ch.1 - ch.16)

* The same processing will be carried out as when All Notes Off is received.
* The Mono/Poly parameter (SCENE PART EDIT: CTRL) will change.

■ System Realtime Message

● Timig Clock

Status

F8H

* Not received when the Sync Mode parameter (SYSTEM:SYNC/TEMPO) is INT.

● Active Sensing

Status

FEH

* When Active Sensing is received, the unit will begin monitoring the intervals of all further messages. While monitoring, if the interval between messages exceeds 420 ms, the same processing will be carried out as when All Sounds Off, All Notes Off and Reset All Controllers are received, and message interval monitoring will be halted.

■ System Exclusive Message

Status	Data byte	Status
F0H	iiH, ddH,, eeH	F7H

F0H: System Exclusive Message status

ii = ID number: an ID number (manufacturer ID) to indicate the manufacturer whose Exclusive message this is. Roland's manufacturer ID is 41H.

ID numbers 7EH and 7FH are extensions of the MIDI standard; Universal Non-realtime Messages (7EH) and Universal Realtime Messages (7FH).

dd,...,ee = data: 00H - 7FH (0 - 127)

F7H: EOX (End Of Exclusive)

The System Exclusive messages received by this device are the Universal Nonrealtime System Exclusive messages and the Universal Realtime System Exclusive messages and the Data Request (RQ1) messages and the Data Set (DT1) messages.

● Universal Non-realtime System Exclusive Messages

o Identity Request Message

Status	Data byte	Status
F0H	7EH, dev, 06H, 01H	F7H

Byte Explanation

F0H Exclusive status

7EH ID number (Universal Non-realtime Message)

dev Device ID (dev: 10H - 1FH, 7FH)

06H Sub ID#1 (General Information)

01H Sub ID#2 (Identity Request)

F7H EOX (End Of Exclusive)

* When this message is received, Identity Reply message will be transmitted.

● Universal Realtime System Exclusive

Messages

o Master Volume

Status	Data byte	Status
F0H	7FH, 7FH, 04H, 01H, 11H, mmH	F7H

Byte Explanation

F0H Exclusive status

7FH ID number (universal realtime message)

7FH Device ID (Broadcast)

04H Sub ID#1 (Device Control)

01H Sub ID#2 (Master Volume)

11H Master Volume lower byte

mmH Master Volume upper byte

F7H EOX (End Of Exclusive)

* The lower byte (11H) of Master Volume will be handled as 00H.

* The Master Volume will change.

o Master Fine Tuning

Status	Data byte	Status
F0H	7FH, 7FH, 04H, 03H, 11H, mmH	F7H
Byte	Explanation	
F0H	Exclusive status	
7FH	ID number (universal realtime message)	
7FH	Device ID (Broadcast)	
04H	Sub ID#1 (Device Control)	
03H	Sub ID#2 (Master Fine Tuning)	
11H	Master Fine Tuning LSB	
mmH	Master Fine Tuning MSB	
F7H	EOX (End Of Exclusive)	
mm,	11: 00 00H - 40 00H - 7F 7FH (-100 - 0 - +99.9 [cents])	
* The MasterTune parameter (SYSTEM: GENERAL) will change.		

o Master Coarse Tuning

Status	Data byte	Status
F0H	7FH, 7FH, 04H, 04H, 11H, mmH	F7
Byte	Explanation	
F0H	Exclusive status	
7FH	ID number (universal realtime message)	
7FH	Device ID (Broadcast)	
04H	Sub ID#1 (Device Control)	
04H	Sub ID#2 (Master Coarse Tuning)	
11H	Master Coarse Tuning LSB	
mmH	Master Coarse Tuning MSB	
F7H	EOX (End Of Exclusive)	
mmH:	28H - 40H - 58H (-24 - 0 - +24 [semitones])	
11H:	ignored (processed as 00H)	
* The MasKeyShift parameter (SYSTEM: GENERAL) will change.		

● Global Parameter Control

o Scale/Octave Tuning Adjust

Status	Data byte	Status
F0H	7EH, 7FH, 08H, 08H, ffH, ggH, hhH, ssH...	F7
Byte	Explanation	
F0H	Exclusive status	
7EH	ID number (Universal Non-realtime Message)	
7FH	Device ID (Broadcast)	
08H	Sub ID#1 (MIDI Tuning Standard)	
08H	Sub ID#2 (scale/octave tuning 1-byte form)	
ffH	Channel/Option byte 1	
bits	0 to 1 = channel 15 to 16	
bit	2 to 6 = Undefined	
ggH	Channel byte 2	
bits	0 to 6 = channel 8 to 14	
hhH	Channel byte 3	
bits	0 to 6 = channel 1 to 7	
ssH	12 byte tuning offset of 12 semitones from C to B	
00H	= -64 [cents]	
40H	= 0 [cents] (equal temperament)	
7FH	= +63 [cents]	
F7H	EOX (End Of Exclusive)	

● Data Transmission

This instrument can use exclusive messages to exchange many varieties of internal settings with other devices.

The model ID of the exclusive messages used by this instrument is 00H 00H 00H 00H 12H.

o Data Request 1 (RQ1)

This message requests the other device to transmit data. The address and size indicate the type and amount of data that is requested.

When a Data Request message is received, if the device is in a state in which it is able to transmit data, and if the address and size are appropriate, the requested data is transmitted as a Data Set 1 (DT1) message. If the conditions are not met, nothing is transmitted.

Status	data byte	Status
F0H	41H, dev, 00H, 00H, 00H, 00H, 12H, 11H, aaH, bbH, F7H	
	ccH, ddH, ssH, ttH, uuH, vvH, sum	
Byte	Explanation	
F0H	Exclusive status	
41H	ID number (Roland)	
dev	device ID (dev: 10H - 1FH, 7FH)	
00H	model ID #1 (JUNO-X)	
00H	model ID #2 (JUNO-X)	
00H	model ID #3 (JUNO-X)	
00H	model ID #4 (JUNO-X)	
12H	model ID #5 (JUNO-X)	
11H	command ID (RQ1)	
aaH	address MSB	
bbH	address	
ccH	address	
ddH	address LSB	
ssH	size MSB	
ttH	size	
uuH	size	
vvH	size LSB	
sum	checksum	
F7H	EOX (End Of Exclusive)	

- * The size of data that can be transmitted at one time is fixed for each type of data. And data requests must be made with a fixed starting address and size. Refer to the address and size given in Parameter Address Map.
- * For the checksum, refer to "How to calculate the checksum".
- * Not received when the Rx Exclusive parameter (SYSTEM: MIDI RX) is OFF.

o Data set 1 (DT1)

This is the message that actually performs data transmission, and is used when you wish to transmit the data.

Status	data byte	Status
F0H	41H, dev, 00H, 00H, 00H, 00H, 12H, 12H, aaH, bbH, F7H ccH, ddH, eeH, ... ffH, sum	

Byte	Explanation
F0H	Exclusive status
41H	ID number (Roland)
dev	device ID (dev: 10H - 1FH, 7FH)
00H	model ID #1 (JUNO-X)
00H	model ID #2 (JUNO-X)
00H	model ID #3 (JUNO-X)
00H	model ID #4 (JUNO-X)
12H	model ID #5 (JUNO-X)
12H	Command ID (DT1)
aaH	address MSB
bbH	address
ccH	address
ddH	address LSB
eeH Data:	the actual data to be sent. Multiple bytes of data are transmitted in order starting from the address.
:	:
ffH	Data
sum	checksum
F7H	EOX (End Of Exclusive)

- * The amount of data that can be transmitted at one time depends on the type of data, and data will be transmitted from the specified starting address and size. Refer to the address and size given in Parameter Address Map.
- * Data larger than 256 bytes will be divided into packets of 256 bytes or less, and each packet will be sent at an interval of about 20 ms.
- * For the checksum, refer to "How to calculate the checksum".
- * Not received when the Rx Exclusive parameter (SYSTEM: MIDI RX) is OFF.

2. Data Transmission

■ Channel Voice Messages

● Note Off

Status	2nd byte	3rd byte
8nH	kkH	vvH

n = MIDI channel number: 0H - FH (ch.1 - ch.16)
kk = Note number: 00H - 7FH (0 - 127)
vv = Note off velocity: 00H - 7FH (0 - 127)

● Note On

Status	2nd byte	3rd byte
9nH	kkH	vvH

n = MIDI channel number: 0H - FH (ch.1 - ch.16)
kk = Note number: 00H - 7FH (0 - 127)
vv = Note on velocity: 01H - 7FH (1 - 127)

● Control Change

- * By selecting a controller number that corresponds to the setting of parameters of controllers (knobs, etc.), this unit can transmit any control change message.

o Bank Select (Controller number 0, 32)

Status	2nd byte	3rd byte
BnH	00H	mmH
BnH	20H	11H

n = MIDI channel number: 0H - FH (ch.1 - ch.16)

mm, 11 = Bank number: 00 00H - 7F 7FH (bank.1 - bank.16384)

- * These messages are transmitted when Scene or Tone is selected. But not transmitted when Tx PC (SYSTEM: MIDI TX) or Tx Bank parameter (SYSTEM: MIDI TX) is OFF.

o Modulation (Controller number 1)

Status	2nd byte	3rd byte
BnH	01H	vvH

n = MIDI channel number: 0H - FH (ch.1 - ch.16)

vv = Modulation depth: 00H - 7FH (0 - 127)

o Breath type (Controller number 2)

Status	2nd byte	3rd byte
BnH	02H	vvH

n = MIDI channel number: 0H - FH (ch.1 - ch.16)

vv = Control value: 00H - 7FH (0 - 127)

o Foot Type (Controller number 4)

Status	2nd byte	3rd byte
BnH	04H	vvH

n = MIDI channel number: 0H - FH (ch.1 - 16)

vv = Control value: 00H - 7FH (0 - 127)

o Portamento Time (Controller number 5)

Status	2nd byte	3rd byte
BnH	05H	vvH

n = MIDI channel number: 0H - FH (ch.1 - ch.16)

vv = Portamento Time: 00H - 7FH (0 - 127)

o Data Entry (Controller number 6, 38)

Status	2nd byte	3rd byte
BnH	06H	mmH
BnH	26H	11H

n = MIDI channel number: 0H - FH (ch.1 - ch.16)

mm, 11 = the value of the parameter specified by RPN/NRPN

mm = MSB, 11 = LSB

o Volume (Controller number 7)

Status	2nd byte	3rd byte
BnH	07H	vvH

n = MIDI channel number: 0H - FH (ch.1 - ch.16)

vv = Volume: 00H - 7FH (0 - 127)

o Panpot (Controller number 10)

Status	2nd byte	3rd byte
BnH	0AH	vvH

n = MIDI channel number: 0H - FH (ch.1 - ch.16)
 vv = Panpot: 00H - 40H - 7FH (Left - Center - Right)

o Expression (Controller number 11)

Status	2nd byte	3rd byte
BnH	0BH	vvH

n = MIDI channel number: 0H - FH (ch.1 - ch.16)
 vv = Expression: 00H - 7FH (0 - 127)

o General Purpose Controller 1 (Controller number 16)

Status	2nd byte	3rd byte
BnH	10H	vvH

n = MIDI channel number: 0H - FH (ch.1 - ch.16)
 vv = Control value: 00H - 7FH (0 - 127)

o General Purpose Controller 2 (Controller number 17)

Status	2nd byte	3rd byte
BnH	11H	vvH

n = MIDI channel number: 0H - FH (ch.1 - ch.16)
 vv = Control value: 00H - 7FH (0 - 127)

o General Purpose Controller 3 (Controller number 18)

Status	2nd byte	3rd byte
BnH	12H	vvH

n = MIDI channel number: 0H - FH (ch.1 - ch.16)
 vv = Control value: 00H - 7FH (0 - 127)

o General Purpose Controller 4 (Controller number 19)

Status	2nd byte	3rd byte
BnH	13H	vvH

n = MIDI channel number: 0H - FH (ch.1 - ch.16)
 vv = Control value: 00H - 7FH (0 - 127)

o Hold 1 (Controller number 64)

Status	2nd byte	3rd byte
BnH	40H	vvH

n = MIDI channel number: 0H - FH (ch.1 - ch.16)
 vv = Control value: 00H - 7FH (0 - 127)
 0-63 = OFF, 64-127 = ON

o Portamento (Controller number 65)

Status	2nd byte	3rd byte
BnH	41H	vvH

n = MIDI channel number: 0H - FH (ch.1 - ch.16)
 vv = Control value: 00H - 7FH (0 - 127)
 0-63 = OFF, 64-127 = ON

o Sostenuto (Controller number 66)

Status	2nd byte	3rd byte
BnH	42H	vvH

n = MIDI channel number: 0H - FH (ch.1 - ch.16)
 vv = Control value: 00H - 7FH (0 - 127)

o Soft (Controller number 67)

Status	2nd byte	3rd byte
BnH	43H	vvH

n = MIDI channel number: 0H - FH (ch.1 - ch.16)
 vv = Control value: 00H - 7FH (0 - 127)

o Legato Foot Switch (Controller number 68)

Status	2nd byte	3rd byte
BnH	44H	vvH

n = MIDI channel number: 0H - FH (ch.1 - ch.16)
 vv = Control value: 00H - 7FH (0 - 127)

o Resonance (Controller number 71)

Status	2nd byte	3rd byte
BnH	47H	vvH

n = MIDI channel number: 0H - FH (ch.1 - ch.16)
 vv = Resonance value (relative change): 00H - 40H - 7FH (-64 - 0 - +63)

o Release Time (Controller number 72)

Status	2nd byte	3rd byte
BnH	48H	vvH

n = MIDI channel number: 0H - FH (ch.1 - ch.16)
 vv = Release Time value (relative change): 00H - 40H - 7FH (-64 - 0 - +63)

o Attack time (Controller number 73)

Status	2nd byte	3rd byte
BnH	49H	vvH

n = MIDI channel number: 0H - FH (ch.1 - ch.16)
 vv = Attack time value (relative change): 00H - 40H - 7FH (-64 - 0 - +63)

o Cutoff (Controller number 74)

Status	2nd byte	3rd byte
BnH	4AH	vvH

n = MIDI channel number: 0H - FH (ch.1 - 16)
 vv = Cutoff value (relative change): 00H - 40H - 7FH (-64 - 0 - +63)

o General Purpose Controller 5 (Controller number 80)

Status	2nd byte	3rd byte
BnH	50H	vvH

n = MIDI channel number: 0H - FH (ch.1 - ch.16)
 vv = Control value: 00H - 7FH (0 - 127)

o General Purpose Controller 6 (Controller number 81)

Status	2nd byte	3rd byte
BnH	51H	vvH

n = MIDI channel number: 0H - FH (ch.1 - ch.16)

vv = Control value: 00H - 7FH (0 - 127)

o General Purpose Controller 7 (Controller number 82)

Status	2nd byte	3rd byte
BnH	52H	vvH

n = MIDI channel number: 0H - FH (ch.1 - ch.16)

vv = Control value: 00H - 7FH (0 - 127)

o General Purpose Controller 8 (Controller number 83)

Status	2nd byte	3rd byte
BnH	53H	vvH

n = MIDI channel number: 0H - FH (ch.1 - ch.16)

vv = Control value: 00H - 7FH (0 - 127)

o Portamento control (Controller number 84)

Status	2nd byte	3rd byte
BnH	54H	kkH

n = MIDI channel number: 0H - FH (ch.1 - ch.16)

kk = source note number: 00H - 7FH (0 - 127)

o General Purpose Effect 1 (Reverb Send Level)

(Controller number 91)

Status	2nd byte	3rd byte
BnH	5BH	vvH

n = MIDI channel number: 0H - FH (ch.1 - ch.16)

vv = Reverb Send Level: 00H - 7FH (0 - 127)

o General Purpose Effect 3 (Chorus Send Level)

(Controller number 93)

Status	2nd byte	3rd byte
BnH	5DH	vvH

n = MIDI channel number: 0H - FH (ch.1 - ch.16)

vv = Chorus Send Level: 00H - 7FH (0 - 127)

o General Controller

Status	2nd byte	3rd byte
BnH	kkH	vvH

n = MIDI channel number: 0H - FH (ch.1 - ch.16)

kk = Controller number: 00H - 77H (0 - 31, 33 - 95)

vv = Control value: 00H - 7FH (0 - 127)

● Program Change

Status	2nd byte
CnH	ppH

n = MIDI channel number: 0H - FH (ch.1 - ch.16)

pp = Program number: 00H - 7FH (prog.1 - prog.128)

* These messages are transmitted when Program or Tone is selected. But not transmitted when Transmit Program Change parameter (SYSTEM: MIDI TX) is OFF.

● Channel Pressure

Status	2nd byte
DnH	vvH

n = MIDI channel number: 0H - FH (ch.1 - ch.16)

vv = Channel Pressure: 00H - 7FH (0 - 127)

● Pitch Bend Change

Status	2nd byte	3rd byte
EnH	11H	mmH

n = MIDI channel number: 0H - FH (ch.1 - ch.16)

mm, 11 = Pitch Bend value: 00 00H - 40 00H - 7F 7FH (-8192 - 0 - +8191)

■ Channel Mode Messages

● MONO (Controller number 126)

Status	2nd byte	3rd byte
BnH	7EH	mmH

n = MIDI channel number: 0H - FH (ch.1 - ch.16)

mm = mono number: 00H - 10H (0 - 16)

● POLY (Controller number 127)

Status	2nd byte	3rd byte
BnH	7FH	00H

n = MIDI channel number: 0H - FH (ch.1 - ch.16)

■ System Realtime Message

● Timing Clock

Status

F8H

* Transmitted when the Sync Mode parameter (SYSTEM:SYNC/TEMPO) is INT.

● Active Sensing

Status

FEH

* This message is transmitted at intervals of approximately 250 msec.

■ System Exclusive Message

Universal Non-realtime System Exclusive Message and Data Set 1 (DT1) are the only System Exclusive messages transmitted by this unit.

● Universal Non-realtime System Exclusive Messages

○ Identity Reply Message

JUNO-X

Status	data byte	Status
F0H	7EH, 7FH, 06H, 02H, 41H, 12H, 04H, 00H, 00H, 00H, 01H, 00H, 01H,	F7H
Byte	Explanation	
F0H	Exclusive status	
7EH	ID number (Universal Non-realtime Message)	
dev	Device ID (dev: 10H - 1FH, 7FH)	
06H	Sub ID#1 (General Information)	
02H	Sub ID#2 (Identity Reply)	
41H	ID number (Roland)	
12H 04H	Device family code	
00H 00H	Device family number code	
00H 01H 00H 01H	Software revision level	
F7H	EOX (End Of Exclusive)	

● Data Transmission

○ Data set 1 (DT1)

Status	data byte	Status
F0H	41H, dev, 00H, 00H, 00H, 00H, 12H, 12H, aaH, bbH, F7H	
	ccH, ddH, eeH, ... ffH, sum,	
Byte	Explanation	
F0H	Exclusive status	
41H	ID number (Roland)	
dev	device ID (dev: 10H - 1FH, 7FH)	
00H	model ID #1 (JUNO-X)	
00H	model ID #2 (JUNO-X)	
00H	model ID #3 (JUNO-X)	
00H	model ID #4 (JUNO-X)	
12H	model ID #5 (JUNO-X)	
12H	Command ID (DT1)	
aaH	Address MSB: upper byte of the starting address of the data to be sent	
bbH	Address: upper middle byte of the starting address of the data to be sent	
ccH	Address: lower middle byte of the starting address of the data to be sent	
ddH	Address LSB: lower byte of the starting address of the data to be sent.	
eeH Data:	the actual data to be sent. Multiple bytes of data are transmitted in order starting from the address.	
:	:	
ffH	Data	
sum	checksum	
F7H	EOX (End Of Exclusive)	

* The size of data that can be transmitted at one time is fixed for each type of data. And data requests must be made with a fixed starting address and size. Refer to the address and size given in Parameter Address Map.

* Data larger than 256 bytes will be divided into packets of 256 bytes or less, and each packet will be sent at an interval of about 20 ms.

3. Parameter Address Map

* Transmission of "#" marked address is divided to some packets. For example, ABH in hexadecimal notation will be divided to 0AH and 0BH, and is sent/received in this order.

JUNO-X (ModelID = 00H 00H 00H 00H 12H)

Start	Address	Description	
00 00 00 00		System	[System]
00 10 00 00		Setup	[Setup]
01 00 00 00		Temporary Scene	[Scene]
02 00 00 00		Temporary Tone ZCore (01)	[ZCore]
02 01 00 00		Temporary Tone ZCore (02)	[ZCore]
:			
02 03 00 00		Temporary Tone ZCore (04)	[ZCore]
02 10 00 00		Temporary Tone Analog Synth Model (01)	[SynMdl]
02 11 00 00		Temporary Tone Analog Synth Model (02)	[SynMdl]
:			
02 13 00 00		Temporary Tone Analog Synth Model (04)	[SynMdl]
02 20 00 00		Temporary Tone RD Piano (01)	[RDTone]
02 40 00 00		Temporary Tone JUNO-X Model (01)	[JunoXMdl]
02 41 00 00		Temporary Tone JUNO-X Model (02)	[JunoXMdl]
:			
02 43 00 00		Temporary Tone JUNO-X Model (04)	[JunoXMdl]
40 00 00 00		User Scene (001)	[Scene]
40 05 00 00		User Scene (002)	[Scene]
:			

49 7B 00 00	User Scene (256)	[Scene]
50 00 00 00	User Tone (001)	[UserTone]
50 02 00 00	User Tone (002)	[UserTone]
:		
53 7E 00 00	User Tone (256)	[UserTone]

* [System]

Offset	Address	Description
00 00 00	System Common	[System Common]
00 01 00	System Control	[System Control]
00 02 00	System Chorus	[Chorus]
00 03 00	System Delay	[Chorus]
00 04 00	System Reverb	[Reverb]
00 05 00	Master EQ	[Master EQ]
00 06 00	Master Comp	[Master Comp]
00 07 00	Model Bank (01) Assign	[Model Bank Assign]
00 08 00	Model Bank (02) Assign	[Model Bank Assign]
:		
00 16 00	Model Bank (16) Assign	[Model Bank Assign]
00 17 00	Button Color Set (01)	[Button Color Set]
00 18 00	Button Color Set (02)	[Button Color Set]
:		
00 20 00	Button Color Set (10)	[Button Color Set]

* [SynMdl]

Offset	Address	Description
00 00 00	Analog Synth Model Parameter	[MDLSYNO]
00 02 00	Analog Synth Model MFX	[MFX]
00 04 00	Analog Synth Model Tone Com	[ToneCom]

* [RDTone]

Offset	Address	Description
00 00 00	RD Tone Name	[ToneName]
00 01 00	RD Piano Sympathetic Resonan	[Sympathetic Resonance]

* [JunoXMdl]

Offset	Address	Description
00 00 00	JUNO-X Model Parameter	[MDLJUNOX]
00 02 00	JUNO-X Model MFX	[MFX]
00 04 00	JUNO-X Model Tone Com	[ToneCom]

* [Scene]

Offset	Address	Description
00 00 00	Scene Common	[Scene Common]
00 10 00	Scene Part (01)	[Scene Part]
00 11 00	Scene Part (02)	[Scene Part]
:		
00 14 00	Scene Part (05)	[Scene Part]
00 20 00	Scene EQ (01)	[Scene EQ]
00 21 00	Scene EQ (02)	[Scene EQ]
:		
00 24 00	Scene EQ (05)	[Scene EQ]
00 30 00	Scene MFX (01)	[MFX]
00 32 00	Scene MFX (02)	[MFX]
:		
00 38 00	Scene MFX (05)	[MFX]
00 40 00	Scene Zone (01)	[Scene Zone]
00 41 00	Scene Zone (02)	[Scene Zone]
:		
00 44 00	Scene Zone (05)	[Scene Zone]

00 50 00	Scene Chorus	[Chorus]
00 51 00	Scene Delay	[Chorus]
00 52 00	Scene Reverb	[Reverb]
00 53 00	Scene Drive	[Drive]
00 54 00	Scene Arpeggio Common	[Arpeggio Common]
00 55 00	Scene Arpeggio Part (01)	[Arpeggio Part]
00 56 00	Scene Arpeggio Part (02)	[Arpeggio Part]
:		
00 59 00	Scene Arpeggio Part (05)	[Arpeggio Part]
00 5A 00	Scene Arpeggio User Pattern (01)	[User Pattern]
00 7A 00	Scene Arpeggio User Pattern (02)	[User Pattern]
:		
01 5A 00	Scene Arpeggio User Pattern (05)	[User Pattern]

* [ZCore]

Offset	Address	Description
00 00 00	Tone Common	[Tone Common]
00 01 00	Tone MFX	[MFX]
00 10 00	Tone PMT	[Tone PMT]
00 20 00	Tone Partial 1	[Tone Partial]
00 21 00	Tone Partial 2	[Tone Partial]
:		
00 23 00	Tone Partial 4	[Tone Partial]
00 24 00	Partial Pitch Env 1	[Partial Pitch Env]
00 25 00	Partial Pitch Env 2	[Partial Pitch Env]
:		
00 27 00	Partial Pitch Env 4	[Partial Pitch Env]
00 28 00	Partial Filter Env 1	[Partial Filter Env]
00 29 00	Partial Filter Env 2	[Partial Filter Env]
:		
00 2B 00	Partial Filter Env 4	[Partial Filter Env]
00 2C 00	Partial Amp Env 1	[Partial Amp Env]
00 2D 00	Partial Amp Env 2	[Partial Amp Env]
:		
00 2F 00	Partial Amp Env 4	[Partial Amp Env]
00 30 00	Partial LFO 1	[Partial LFO]
00 32 00	Partial LFO 2	[Partial LFO]
:		
00 36 00	Partial LFO 4	[Partial LFO]
00 38 00	Partial EQ 1	[Partial EQ]
00 39 00	Partial EQ 2	[Partial EQ]
:		
00 3B 00	Partial EQ 4	[Partial EQ]
00 3C 00	Synth Common	[Tone Synth Common]
00 3D 00	Synth PMT	[Tone Synth PMT]
00 3E 00	Synth Partial 1	[Tone Synth Partial]
00 3F 00	Synth Partial 2	[Tone Synth Partial]
:		
00 41 00	Synth Partial 4	[Tone Synth Partial]

* [UserTone]

Offset	Address	Description
00 00 00	data 1	[USRDATA1BLK]
00 02 00	data 2	[USRDATA2BLK]
:		
00 1E 00	data 16	[USRDATA16BLK]

* [Setup]

Offset	Address	Description
00 00	0aaa aaaa	Scene BS MSB (CC# 0) (0 - 127)
00 01	0aaa aaaa	Scene BS LSB (CC# 32) (0 - 127)
00 02	0aaa aaaa	Scene PC (PC) (0 - 127)
00 00 00 03	Total Size	

* [System Common]

Offset	Address	Description
--------	---------	-------------

		PANEL INC,CHO SW,REV SW,DLY SW,ARP SW,ARP HOLD, DETECT KEYS,DETECT BEAT,UNISON SW,BEND MODE, AUTO TUNING,TAP TEMPO,START/STOP,DRV SW				CHO:CC93,CELST:CC94,PHASR:CC95,AFT,BEND DOWN, BEND UP,CHO LEVEL,REV LEVEL,DLY LEVEL,ARP SHUFFLE, ARP DURATION,PART FADE1,PART FADE2,LEVEL P-1, LEVEL P-2,LEVEL P-3,LEVEL P-4,LEVEL P-R,AGE	
00 07	0aaa aaaa	Assignnable Button(3) Function (0 - 116)	00 14	0aaa aaaa	Assignnable Slider(2) Function (0 - 111)		
		OFF,MOD:CC01, BRETH:CC02,CC03,FOOT:CC04,PTIME:CC05,DENT:CC06, VOL:CC07,BAL:CC08,CC09,PAN:CC10,EXP:CC11, CC12,CC13,CC14,CC15,GEN-1:CC16,GEN-2:CC17, GEN-3:CC18,GEN-4:CC19,CC20,CC21,CC22,CC23, CC24,CC25,CC26,CC27,CC28,CC29,CC30,CC31,OFF:CC32, CC33,CC34,CC35,CC36,CC37,DENT:CC38,CC39,CC40, CC41,CC42,CC43,CC44,CC45,CC46,CC47,CC48,CC49,CC50, CC51,CC52,CC53,CC54,CC55,CC56,CC57,CC58,CC59,CC60, CC61,CC62,CC63,HOLD1:CC64,PORTA:CC65,SOSTE:CC66, SOFT:CC67,LEGAT:CC68,HOLD2:CC69,CC70,RESO:CC71, RELES:CC72,ATTAK:CC73,CUTOF:CC74,DECAY:CC75, VRATE:CC76,VDPATH:CC77,VDLY:CC78,CC79,GEN-5:CC80, GEN-6:CC81,GEN-7:CC82,GEN-8:CC83,PCTRL:CC84, CC85,CC86,CC87,CC88,CC89,CC90,REV:CC91,TREM:CC92, CHO:CC93,CELST:CC94,PHASR:CC95,AFT,MONO/POLY, SCENE DOWN,SCENE UP,TONE DOWN,TONE UP,PANEL DEC, PANEL INC,CHO SW,REV SW,DLY SW,ARP SW,ARP HOLD, DETECT KEYS,DETECT BEAT,UNISON SW,BEND MODE, AUTO TUNING,TAP TEMPO,START/STOP,DRV SW			OFF,MOD:CC01, BRETH:CC02,CC03,FOOT:CC04,PTIME:CC05,DENT:CC06, VOL:CC07,BAL:CC08,CC09,PAN:CC10,EXP:CC11, CC12,CC13,CC14,CC15,GEN-1:CC16,GEN-2:CC17, GEN-3:CC18,GEN-4:CC19,CC20,CC21,CC22,CC23, CC24,CC25,CC26,CC27,CC28,CC29,CC30,CC31,OFF:CC32, CC33,CC34,CC35,CC36,CC37,DENT:CC38,CC39,CC40, CC41,CC42,CC43,CC44,CC45,CC46,CC47,CC48,CC49,CC50, CC51,CC52,CC53,CC54,CC55,CC56,CC57,CC58,CC59,CC60, CC61,CC62,CC63,HOLD1:CC64,PORTA:CC65,SOSTE:CC66, SOFT:CC67,LEGAT:CC68,HOLD2:CC69,CC70,RESO:CC71, RELES:CC72,ATTAK:CC73,CUTOF:CC74,DECAY:CC75, VRATE:CC76,VDPATH:CC77,VDLY:CC78,CC79,GEN-5:CC80, GEN-6:CC81,GEN-7:CC82,GEN-8:CC83,PCTRL:CC84, CC85,CC86,CC87,CC88,CC89,CC90,REV:CC91,TREM:CC92, CHO:CC93,CELST:CC94,PHASR:CC95,AFT,BEND DOWN, BEND UP,CHO LEVEL,REV LEVEL,DLY LEVEL,ARP SHUFFLE, ARP DURATION,PART FADE1,PART FADE2,LEVEL P-1, LEVEL P-2,LEVEL P-3,LEVEL P-4,LEVEL P-R,AGE		
00 08	0000 000a	Assignnable Button(1) Mode (0 - 1)	00 15	0000 000a	Reserved (0 - 0)		
		LATCH, MOMENTARY	00 16	0aaa aaaa	Reserved (0 - 0)		
00 09	0000 000a	Assignnable Button(2) Mode (0 - 1)	00 17	0000 000a	Reserved (0 - 0)		
		LATCH, MOMENTARY	00 18	0aaa aaaa	Reserved (0 - 0)		
00 0A	0000 000a	Assignnable Button(3) Mode (0 - 1)	00 19	0000 0aaa	Part Button Assign [4]-[8] (0 - 6)		
		LATCH, MOMENTARY			NO ASSIGN, PART SELECT, PART SEL + KBD SW, KEYBOARD SW, PART SW, ARP PART SW, EFFECT SW		
00 0B	0000 000a	Assignnable Button Source (0 - 1)	00 1A	0000 0aaa	Part Button Assign SHIFT+[4]-[8] (0 - 6)		
		SCENE, SYS			NO ASSIGN, PART SELECT, PART SEL + KBD SW, KEYBOARD SW, PART SW, ARP PART SW, EFFECT SW		
00 0C	0000 000a	Hold Pedal Source (0 - 1)			NO ASSIGN, PART SELECT, PART SEL + KBD SW, KEYBOARD SW, PART SW, ARP PART SW, EFFECT SW		
		SCENE, SYS			KEYBOARD SW, PART SW, ARP PART SW, EFFECT SW		
00 0D	0aaa aaaa	Hold Pedal Function (0 - 116)	00 1B	0000 0aaa	Part Button Assign [9]-[13] (0 - 6)		
		OFF,MOD:CC01, BRETH:CC02,CC03,FOOT:CC04,PTIME:CC05,DENT:CC06, VOL:CC07,BAL:CC08,CC09,PAN:CC10,EXP:CC11, CC12,CC13,CC14,CC15,GEN-1:CC16,GEN-2:CC17, GEN-3:CC18,GEN-4:CC19,CC20,CC21,CC22,CC23, CC24,CC25,CC26,CC27,CC28,CC29,CC30,CC31,OFF:CC32, CC33,CC34,CC35,CC36,CC37,DENT:CC38,CC39,CC40, CC41,CC42,CC43,CC44,CC45,CC46,CC47,CC48,CC49,CC50, CC51,CC52,CC53,CC54,CC55,CC56,CC57,CC58,CC59,CC60, CC61,CC62,CC63,HOLD1:CC64,PORTA:CC65,SOSTE:CC66, SOFT:CC67,LEGAT:CC68,HOLD2:CC69,CC70,RESO:CC71, RELES:CC72,ATTAK:CC73,CUTOF:CC74,DECAY:CC75, VRATE:CC76,VDPATH:CC77,VDLY:CC78,CC79,GEN-5:CC80, GEN-6:CC81,GEN-7:CC82,GEN-8:CC83,PCTRL:CC84, CC85,CC86,CC87,CC88,CC89,CC90,REV:CC91,TREM:CC92, CHO:CC93,CELST:CC94,PHASR:CC95,AFT,MONO/POLY, SCENE DOWN,SCENE UP,TONE DOWN,TONE UP,PANEL DEC, PANEL INC,CHO SW,REV SW,DLY SW,ARP SW,ARP HOLD, DETECT KEYS,DETECT BEAT,UNISON SW,BEND MODE, AUTO TUNING,TAP TEMPO,START/STOP,DRV SW			NO ASSIGN, PART SELECT, PART SEL + KBD SW, KEYBOARD SW, PART SW, ARP PART SW, EFFECT SW		
			00 1C	0000 0aaa	Part Button Assign SHIFT+[9]-[13] (0 - 6)		
			00 1D	0000 0aaa	Reserved (0 - 0)		
			00 1E	0000 0aaa	Reserved (0 - 0)		
			00 1F	0aaa aaaa	Aftertouch Sens (0 - 100)		
			00 00 00 20	Total Size			

* [Button Color Set]

Offset	Address	Description
00 00	000a aaaa	Button Color(1) (0 - 16) OFF, ORANGE, ORANGE BLINK, YELLOW, YELLOW BLINK, WHITE, WHITE BLINK, GREEN, GREEN BLINK, SKYBLUE, SKYBLUE BLINK, RED, RED BLINK, VIOLET, VIOLET BLINK, BLUE, BLUE BLINK
00 01	000a aaaa	Button Color(2) (0 - 16) OFF, ORANGE, ORANGE BLINK, YELLOW, YELLOW BLINK, WHITE, WHITE BLINK, GREEN, GREEN BLINK, SKYBLUE, SKYBLUE BLINK, RED, RED BLINK, VIOLET, VIOLET BLINK, BLUE, BLUE BLINK
00 02	000a aaaa	Button Color(3) (0 - 16) OFF, ORANGE, ORANGE BLINK, YELLOW, YELLOW BLINK, WHITE, WHITE BLINK, GREEN, GREEN BLINK, SKYBLUE, SKYBLUE BLINK, RED, RED BLINK, VIOLET, VIOLET BLINK, BLUE, BLUE BLINK
00 03	000a aaaa	Button Color(4) (0 - 16) OFF, ORANGE, ORANGE BLINK, YELLOW, YELLOW BLINK, WHITE, WHITE BLINK, GREEN, GREEN BLINK, SKYBLUE, SKYBLUE BLINK, RED, RED BLINK, VIOLET, VIOLET BLINK, BLUE, BLUE BLINK
00 04	000a aaaa	Button Color(5) (0 - 16) OFF, ORANGE, ORANGE BLINK, YELLOW, YELLOW BLINK, WHITE, WHITE BLINK, GREEN, GREEN BLINK, SKYBLUE, SKYBLUE BLINK, RED, RED BLINK, VIOLET, VIOLET BLINK, BLUE, BLUE BLINK
00 05	000a aaaa	Button Color(6) (0 - 16) OFF, ORANGE, ORANGE BLINK, YELLOW, YELLOW BLINK, WHITE, WHITE BLINK, GREEN, GREEN BLINK, SKYBLUE, SKYBLUE BLINK, RED, RED BLINK, VIOLET, VIOLET BLINK, BLUE, BLUE BLINK
00 06	000a aaaa	Button Color(7) (0 - 16) OFF, ORANGE, ORANGE BLINK, YELLOW, YELLOW BLINK, WHITE, WHITE BLINK, GREEN, GREEN BLINK, SKYBLUE, SKYBLUE BLINK, RED, RED BLINK, VIOLET, VIOLET BLINK, BLUE, BLUE BLINK
00 07	000a aaaa	Button Color(8) (0 - 16) OFF, ORANGE, ORANGE BLINK, YELLOW, YELLOW BLINK, WHITE, WHITE BLINK, GREEN, GREEN BLINK, SKYBLUE, SKYBLUE BLINK, RED, RED BLINK, VIOLET, VIOLET BLINK, BLUE, BLUE BLINK
00 08	000a aaaa	Button Color(9) (0 - 16) OFF, ORANGE, ORANGE BLINK, YELLOW, YELLOW BLINK, WHITE, WHITE BLINK, GREEN, GREEN BLINK, SKYBLUE, SKYBLUE BLINK, RED, RED BLINK, VIOLET, VIOLET BLINK, BLUE, BLUE BLINK
00 09	000a aaaa	Button Color(10) (0 - 16) OFF, ORANGE, ORANGE BLINK, YELLOW, YELLOW BLINK,

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		CC79, GEN-5: CC80, GEN-6: CC81, GEN-7: CC82, GEN-8: CC83, PCTRL: CC84, CC85, CC86, CC87, CC88, CC89, CC90, REV: CC91, TREM: CC92, CHO: CC93, CELST: CC94, PHASR: CC95, BEND, AFT				74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, TONE	
00 24	Oaaa aaaa	Control Source (2)	(0 - 96)	00 0F	Oaaa aaaa	Part Cutoff Offset (CC#74)	(0 - 127)
		OFF, MOD: CC01, BRETH: CC02, CC03, FOOT: CC04, PTIME: CC05, DENT: CC06, VOL: CC07, BAL: CC08, CC09, PAN: CC10, EXP: CC11, CC12, CC13, CC14, CC15, GEN-1: CC16, GEN-2: CC17, GEN-3: CC18, GEN-4: CC19, CC20, CC21, CC22, CC23, CC24, CC25, CC26, CC27, CC28, CC29, CC30, CC31, CC33, CC34, CC35, CC36, CC37, DENT: CC38, CC39, CC40, CC41, CC42, CC43, CC44, CC45, CC46, CC47, CC48, CC49, CC50, CC51, CC52, CC53, CC54, CC55, CC56, CC57, CC58, CC59, CC60, CC61, CC62, CC63, HOLD1: CC64, PORTA: CC65, SOSTE: CC66, SOFT: CC67, LEGAT: CC68, HOLD2: CC69, CC70, RESO: CC71, RELES: CC72, ATTAK: CC73, CUTOF: CC74, DECAY: CC75, VRATE: CC76, VDPH: CC77, VDLY: CC78, CC79, GEN-5: CC80, GEN-6: CC81, GEN-7: CC82, GEN-8: CC83, PCTRL: CC84, CC85, CC86, CC87, CC88, CC89, CC90, REV: CC91, TREM: CC92, CHO: CC93, CELST: CC94, PHASR: CC95, BEND, AFT		00 10	Oaaa aaaa	Part Resonance Offset (CC#71)	(0 - 127)
				00 11	Oaaa aaaa	Part Attack Time Offset (CC#73)	(0 - 127)
				00 12	Oaaa aaaa	Part Decay Time Offset (CC#75)	(0 - 127)
				00 13	Oaaa aaaa	Part Release Time Offset (CC#72)	(0 - 127)
				00 14	Oaaa aaaa	Part Vibrato Rate (CC#76)	(0 - 127)
				00 15	Oaaa aaaa	Part Vibrato Depth (CC#77)	(0 - 127)
00 25	Oaaa aaaa	Control Source (3)	(0 - 96)	00 16	Oaaa aaaa	Part Vibrato Delay (CC#78)	(0 - 127)
		OFF, MOD: CC01, BRETH: CC02, CC03, FOOT: CC04, PTIME: CC05, DENT: CC06, VOL: CC07, BAL: CC08, CC09, PAN: CC10, EXP: CC11, CC12, CC13, CC14, CC15, GEN-1: CC16, GEN-2: CC17, GEN-3: CC18, GEN-4: CC19, CC20, CC21, CC22, CC23, CC24, CC25, CC26, CC27, CC28, CC29, CC30, CC31, CC33, CC34, CC35, CC36, CC37, DENT: CC38, CC39, CC40, CC41, CC42, CC43, CC44, CC45, CC46, CC47, CC48, CC49, CC50, CC51, CC52, CC53, CC54, CC55, CC56, CC57, CC58, CC59, CC60, CC61, CC62, CC63, HOLD1: CC64, PORTA: CC65, SOSTE: CC66, SOFT: CC67, LEGAT: CC68, HOLD2: CC69, CC70, RESO: CC71, RELES: CC72, ATTAK: CC73, CUTOF: CC74, DECAY: CC75, VRATE: CC76, VDPH: CC77, VDLY: CC78, CC79, GEN-5: CC80, GEN-6: CC81, GEN-7: CC82, GEN-8: CC83, PCTRL: CC84, CC85, CC86, CC87, CC88, CC89, CC90, REV: CC91, TREM: CC92, CHO: CC93, CELST: CC94, PHASR: CC95, BEND, AFT		00 17	0000 00aa	Unison Switch	(0 - 2)
						OFF, ON, TONE	
				00 18	0000 00aa	Bend Mode	(0 - 2)
				00 19	0000 000a	Follow Tone MFX	(0 - 1)
				00 1A	0000 000a	Output Assign	(0 - 1)
						THRU, DRIVE	
				00 1B	Oaaa aaaa	Part Chorus Send Level (CC#93)	(0 - 127)
				00 1C	Oaaa aaaa	Part Reverb Send Level (CC#91)	(0 - 127)
				00 1D	Oaaa aaaa	Part Delay Send Level	(0 - 127)
				00 1E	Oaaa aaaa	Part Octave Shift	(61 - 67)
						-3 - 3	
00 26	Oaaa aaaa	Control Source (4)	(0 - 96)	00 1F	Oaaa aaaa	Part Velocity Sens Offset	(1 - 127)
		OFF, MOD: CC01, BRETH: CC02, CC03, FOOT: CC04, PTIME: CC05, DENT: CC06, VOL: CC07, BAL: CC08, CC09, PAN: CC10, EXP: CC11, CC12, CC13, CC14, CC15, GEN-1: CC16, GEN-2: CC17, GEN-3: CC18, GEN-4: CC19, CC20, CC21, CC22, CC23, CC24, CC25, CC26, CC27, CC28, CC29, CC30, CC31, CC33, CC34, CC35, CC36, CC37, DENT: CC38, CC39, CC40, CC41, CC42, CC43, CC44, CC45, CC46, CC47, CC48, CC49, CC50, CC51, CC52, CC53, CC54, CC55, CC56, CC57, CC58, CC59, CC60, CC61, CC62, CC63, HOLD1: CC64, PORTA: CC65, SOSTE: CC66, SOFT: CC67, LEGAT: CC68, HOLD2: CC69, CC70, RESO: CC71, RELES: CC72, ATTAK: CC73, CUTOF: CC74, DECAY: CC75, VRATE: CC76, VDPH: CC77, VDLY: CC78, CC79, GEN-5: CC80, GEN-6: CC81, GEN-7: CC82, GEN-8: CC83, PCTRL: CC84, CC85, CC86, CC87, CC88, CC89, CC90, REV: CC91, TREM: CC92, CHO: CC93, CELST: CC94, PHASR: CC95, BEND, AFT		00 20	0000 00aa	Voice Assign Mode	(0 - 2)
						SINGLE, LIMIT, FULL	
				00 21	0000 aaaa	Part Scale Tune Type	(0 - 8)
						CUSTOM, EQUAL, JUST-MAJ, JUST-MIN, PYTHAGORE, KIRNBERGE, MEANTONE, WERCKMEIS, ARABIC	
				00 22	0000 aaaa	Part Scale Tune Key	(0 - 11)
						C, C#, D, D#, E, F, F#, G, G#, A, A#, B	
				00 23	Oaaa aaaa	Part Scale Tune for C	(0 - 127)
				00 24	Oaaa aaaa	Part Scale Tune for C#	(0 - 127)
				00 25	Oaaa aaaa	Part Scale Tune for D	(0 - 127)
				00 26	Oaaa aaaa	Part Scale Tune for D#	(0 - 127)
00 27	0000 aaaa	Voice Reserve Layer (1)	(0 - 10)	00 27	Oaaa aaaa	Part Scale Tune for E	(0 - 127)
00 28	0000 aaaa	Voice Reserve Layer (2)	(0 - 10)	00 28	Oaaa aaaa	Part Scale Tune for F	(0 - 127)
00 29	0000 aaaa	Voice Reserve Layer (3)	(0 - 10)			-64 - 63	
00 2A	0000 aaaa	Voice Reserve Layer (4)	(0 - 10)	00 29	Oaaa aaaa	Part Scale Tune for F#	(0 - 127)
00 2B	0000 aaaa	Voice Reserve Layer (5)	(0 - 10)			-64 - 63	
00 2C	0000 0aaa	Current Part	(0 - 4)	00 2A	Oaaa aaaa	Part Scale Tune for G	(0 - 127)
			1 - 5			-64 - 63	
00 2D	Oaaa aaaa	Part Crossfade Position	(0 - 127)	00 2B	Oaaa aaaa	Part Scale Tune for G#	(0 - 127)
			0 - 127			-64 - 63	
-----				00 2C	Oaaa aaaa	Part Scale Tune for A	(0 - 127)
00 00 00 2E Total Size						-64 - 63	
-----				00 2D	Oaaa aaaa	Part Scale Tune for A#	(0 - 127)
						-64 - 63	
* [Scene Part]				00 2E	Oaaa aaaa	Part Scale Tune for B	(0 - 127)
						-64 - 63	
Offset	Address	Description		00 2F	0000 000a	Receive Program Change	(0 - 1)
						OFF, ON	
00 00	0000 000a	Part Switch	(0 - 1)	00 30	0000 000a	Receive Bank Select	(0 - 1)
		OFF, ON				OFF, ON	
00 01	0000 000a	Mute Switch	(0 - 1)	00 31	0000 000a	Receive Pitch Bend	(0 - 1)
		OFF, MUTE				OFF, ON	
00 02	Oaaa aaaa	Tone Bank Select MSB (CC#0)	(0 - 127)	00 32	0000 000a	Receive Poly Key Pressure	(0 - 1)
00 03	Oaaa aaaa	Tone Bank Select LSB (CC#32)	(0 - 127)			OFF, ON	
00 04	Oaaa aaaa	Tone Program Change (PC)	(0 - 127)	00 33	0000 000a	Receive Channel Pressure	(0 - 1)
00 05	Oaaa aaaa	Part Level (CC#7)	(0 - 127)			OFF, ON	
00 06	Oaaa aaaa	Part Pan (CC#10)	(0 - 127)	00 34	0000 000a	Receive Modulation	(0 - 1)
		L64 - 63R				OFF, ON	
00 07	Oaaa aaaa	Part Coarse Tune (RPN#2)	(16 - 112)	00 35	0000 000a	Receive Volume	(0 - 1)
		-48 - 48				OFF, ON	
00 08	Oaaa aaaa	Part Fine Tune (RPN#1)	(14 - 114)	00 36	0000 000a	Receive Pan	(0 - 1)
		-50 - 50				OFF, ON	
00 09	0000 00aa	Part Mono/Poly	(0 - 2)	00 37	0000 000a	Receive Expression	(0 - 1)
		MONO, POLY, TONE				OFF, ON	
00 0A	0000 00aa	Part Legato Switch	(0 - 2)	00 38	0000 000a	Receive Hold-1	(0 - 1)
		OFF, ON, TONE				OFF, ON	
00 0B	000a aaaa	Part Bend Range	(0 - 25)	00 39	0000 000a	Keyboard Velocity Mode	(0 - 1)
		0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, TONE		00 3A	Oaaa aaaa	Keyboard Fixed Velocity	(1 - 127)
00 0C	0000 00aa	Part Portamento Switch (CC#65)	(0 - 2)	00 3B	0000 0aaa	Velocity Curve Type	(0 - 4)
		OFF, ON, TONE				OFF, 1, 2, 3, 4	
#	00 0D	0000 aaaa		00 3C	Oaaa aaaa	Part Amp Attack Time Offset	(0 - 127)
00 0E	0000 bbbb	Part Portamento Time (CC#5)	(0 - 128)			-64 - 63	
		0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, TONE		00 3D	Oaaa aaaa	Part Amp Decay Time Offset	(0 - 127)
						-64 - 63	
				00 3E	Oaaa aaaa	Part Amp Sustain Level Offset	(0 - 127)
						-64 - 63	
				00 3F	Oaaa aaaa	Part Amp Release Time Offset	(0 - 127)
						-64 - 63	

00 40	0aaa aaaa	Part Filter Keyfollow Offset	(0 - 127)	00 11	0000 000a	Reserved	(0 - 0)
00 41	0aaa aaaa	Part Filter Envelope Depth Offset	-64 - 63	00 12	0000 00aa	Tx Mode	(0 - 2) ON, OFF, MKB
00 42	0aaa aaaa	Part Filter Attack Time Offset	(0 - 127)	00 13	0000 aaaa	Master Keyboard Tx CH	(0 - 15) 1 - 16
00 43	0aaa aaaa	Part Filter Decay Time Offset	(0 - 127)	00 14	0000 aaaa		
00 44	0aaa aaaa	Part Filter Sustain Level Offset	-64 - 63	00 15	0000 bbbb	Master Keyboard Bank MSB	(0 - 128) OFF, 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127
00 45	0aaa aaaa	Part Filter Release Time Offset	(0 - 127)				
00 46	0aaa aaaa	Part Pitch Envelope Depth Offset	(0 - 127)				
00 47	0aaa aaaa	Part Pitch Attack Time Offset	(0 - 127)				
00 48	0aaa aaaa	Part Pitch Decay Time Offset	(0 - 127)	00 16	0000 aaaa		
00 49	0aaa aaaa	Part Pitch Sustain Level Offset	-64 - 63	00 17	0000 bbbb	Master Keyboard Bank LSB	(0 - 128) OFF, 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127
00 4A	0aaa aaaa	Part Pitch Release Time Offset	(0 - 127)				
00 4B	0aaa aaaa	Part Filter LFO Depth	-64 - 63				
00 4C	0aaa aaaa	Part Amp LFO Depth	(0 - 127)				
00 4D	0aaa aaaa	Part Pitch LFO Depth	-64 - 63	00 18	0000 aaaa		
00 4E	0000 aaaa	Receive Channel	(0 - 15) 1 - 16	00 19	0000 bbbb	Master Keyboard Program Change	(0 - 128) OFF, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128
00 4F	0000 000a	Panel Edit Switch	(0 - 1) OFF, ON				
00 00 00 50	Total Size						

* [Scene EQ]

Offset	Address	Description
00 00	0aaa aaaa	EQ Input Gain (40 - 88) -24 - +24 [dB]
00 01	0aaa aaaa	EQ Low Gain (40 - 88) -24 - +24 [dB]
00 02	0aaa aaaa	EQ Mid Gain (40 - 88) -24 - +24 [dB]
00 03	0aaa aaaa	EQ High Gain (40 - 88) -24 - +24 [dB]
00 04	000a aaaa	EQ Low Frequency (0 - 29) 20, 25, 31, 40, 50, 63, 80, 100, 125, 160, 200, 250, 315, 400, 500, 630, 800, 1000, 1250, 1600, 2000, 2500, 3150, 4000, 5000, 6300, 8000, 10000, 12600, 16000 [Hz]
00 05	000a aaaa	EQ Mid Frequency (0 - 29) 20, 25, 31, 40, 50, 63, 80, 100, 125, 160, 200, 250, 315, 400, 500, 630, 800, 1000, 1250, 1600, 2000, 2500, 3150, 4000, 5000, 6300, 8000, 10000, 12600, 16000 [Hz]
00 06	000a aaaa	EQ High Frequency (0 - 29) 20, 25, 31, 40, 50, 63, 80, 100, 125, 160, 200, 250, 315, 400, 500, 630, 800, 1000, 1250, 1600, 2000, 2500, 3150, 4000, 5000, 6300, 8000, 10000, 12600, 16000 [Hz]
00 07	0000 0aaa	EQ Mid Q (0 - 5) 0.5, 1.0, 2.0, 4.0, 8.0, 16.0
00 08	0000 000a	Switch (0 - 1) OFF, ON
00 00 00 09	Total Size	

* [Scene Zone]

Offset	Address	Description
00 00	0000 000a	Keyboard Switch (0 - 1) OFF, ON
00 01	0aaa aaaa	Keyboard Range Lower (0 - 127)
00 02	0aaa aaaa	Keyboard Range Upper (0 - 127)
00 03	0aaa aaaa	Keyboard Fade Width Lower (0 - 127)
00 04	0aaa aaaa	Keyboard Fade Width Upper (0 - 127)
00 05	0aaa aaaa	Velocity Range Lower (1 - 127)
00 06	0aaa aaaa	Velocity Range Upper (1 - 127)
00 07	0aaa aaaa	Velocity Fade Width Lower (0 - 127)
00 08	0aaa aaaa	Velocity Fade Width Upper (0 - 127)
00 09	0000 000a	Rx Assignable Button(1) (0 - 1) OFF, ON
00 0A	0000 000a	Rx Assignable Button(2) (0 - 1) OFF, ON
00 0B	0000 000a	Rx Assignable Button(3) (0 - 1) OFF, ON
00 0C	0000 000a	RX Hold Pedal (0 - 1) OFF, ON
00 0D	0000 000a	Rx Control Pedal (0 - 1) OFF, ON
00 0E	0000 000a	Rx Assignable Slider(1) (0 - 1) OFF, ON
00 0F	0000 000a	Rx Assignable Slider(2) (0 - 1) OFF, ON
00 10	0000 000a	Reserved (0 - 0)

00 1A	0000 aaaa		
00 1B	0000 bbbb	Master Keyboard Volume	(0 - 128) OFF, 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127
00 00 00 1C	Total Size		

* [Arpeggio Common]

Offset	Address	Description
00 00	0000 000a	Master Switch (0 - 1) OFF, ON
00 01	0000 000a	Master Hold Switch (0 - 1) OFF, ON
00 02	0000 000a	Keys Detect (0 - 1) OFF, ON
00 03	0000 000a	Beat Detect (0 - 1) OFF, ON
00 04	0000 aaaa	Beat Detect Sens (1 - 10) 1 - 10
00 05	0aaa aaaa	G-Shuffle Rate (0 - 100) [%] 0 - 100 [%]
00 06	0aaa aaaa	G-Duration (14 - 114) -50 - 50
00 07	0000 0000	Reserved (0 - 0)
00 08	0000 aaaa	
00 09	0000 bbbb	
00 0A	0000 cccc	
00 0B	0000 dddd	Original Type No. (0 - 65) 0 - 65
00 0C	0000 aaaa	
00 0D	0000 bbbb	
00 0E	0000 cccc	
00 0F	0000 dddd	Original Rhythm No. (0 - 65) 0 - 65
00 10	0aaa aaaa	Keyboard Input Range Lower (0 - 127) C-1 - G9
00 11	0aaa aaaa	Keyboard Input Range Upper (0 - 127) C-1 - G9
00 00 00 12	Total Size	

* [Arpeggio Part]

Offset	Address	Description
00 00	0000 000a	Part Switch (0 - 1) OFF, ON
00 01	0000 aaaa	

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	00 02	0000 bbbb			00 2F	0000 bbbb	Arp Variation	(0 - 255)
	00 03	0000 cccc						1 - 256
	00 04	0000 dddd	Reserved	(0 - 0)	00 30	0000 000a	Arp Step Switch	(0 - 1)
	00 05	0aaa aaaa	Duration	(0 - 100)				OFF, ON
				0 - 100 [%]	#	00 31	0000 aaaa	
	00 06	0aaa aaaa	Shuffle Rate	(0 - 100)		00 32	0000 bbbb	Probability Style
				0 - 100 [%]		00 33	0aaa aaaa	Probability Amount
	00 07	0aaa aaaa	Octave Range	(61 - 67)				0 - 100 [%]
				-3 - 3		00 34	0000 000a	Probability Amount Direction
	00 08	0aaa aaaa	Transpose	(28 - 100)				(0 - 1)
				-36 - 36		00 35	0000 000a	Probability Amount Auto
	00 09	0000 000a	Part Hold Switch	(0 - 1)				OFF, ON
				OFF, ON		00 36	0aaa aaaa	Probability Velocity Amount
	00 0A	0000 0aaa	Motif	(0 - 7)				0 - 100 [%]
				UP, DOWN,	#	00 37	0000 aaaa	
				UP&DOWN, RANDOM, NOTE ORDER, RHYTHM, PHRASE, AUTO		00 38	0000 bbbb	
	00 0B	0000 000a	Shuffle Resolution	(0 - 1)		00 39	0000 cccc	
				16TH, 8TH		00 3A	0000 dddd	Reserved
	00 0C	0000 0aaa	Grid Note	(0 - 5)	#	00 3B	0000 aaaa	
				4th, 8th, 8th_3, 16th, 16th_3, 32nd		00 3C	0000 bbbb	Probability Grid Offset
	00 0D	0aaa aaaa	End of Grid	(1 - 63)				(-63 - 64)
				1 - 63				-63 - 64
	00 0E	0aaa aaaa	Velocity	(0 - 127)		00 00 00 3D	Total Size	
				REAL, 1, 2, 3, 4, 5, 6, 7,				
				8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24,				
				25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40,				
				41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56,				
				57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72,				
				73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88,				
				89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103,				
				104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115,				
				116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127				
#	00 0F	0000 aaaa						
	00 10	0000 bbbb	Grid Offset	(-63 - 64)				
				-63 - 64	#	00 02	0000 aaaa	
	00 11	0000 000a	Grid Sync	(0 - 1)		00 03	0000 bbbb	[01] Value Step.01
				OFF, ON				(0 - 129)
#	00 12	0000 aaaa						0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11,
	00 13	0000 bbbb	Offset Velocity	(-127 - 127)				12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27,
				-127 - 127				28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43,
								44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59,
	00 14	0aaa aaaa	Ducking Rate	(0 - 100)				60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75,
				0 - 100				76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91,
#	00 15	0000 aaaa						92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105,
	00 16	0000 bbbb	Ducking Note	(0 - 128)				106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117,
				Any, 0, 1, 2, 3, 4, 5, 6, 7,				118, 119, 120, 121, 122, 123, 124, 125, 126, 127, OFF, TIE
				8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24,	#	00 04	0000 aaaa	
				25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40,		00 05	0000 bbbb	[01] Value Step.02
				41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56,				(0 - 129)
				57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72,				0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11,
				73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88,				12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27,
				89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103,				28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43,
				104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115,				44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59,
				116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127				60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75,
								76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91,
	00 17	0000 0aaa	Ducking Part	(0 - 5)				92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105,
				OFF, PART 1, PART 2, PART 3, PART 4, PART R				106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117,
	00 18	0aaa aaaa	Key Range Low	(0 - 127)				118, 119, 120, 121, 122, 123, 124, 125, 126, 127, OFF, TIE
				0 - 127	#	00 06	0000 aaaa	
	00 19	0000 aaaa	Key Range Octave	(0 - 12)		00 07	0000 bbbb	[01] Value Step.03
				OFF, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12				(0 - 129)
								0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11,
	00 1A	0000 000a	Receive Switch	(0 - 1)				12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27,
				OFF, ON				28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43,
	00 1B	0000 000a	Note Off Mode	(0 - 1)				44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59,
				NORMAL, IMMEDIATE				60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75,
	00 1C	0000 00aa	Pattern Change Timing	(0 - 3)				76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91,
				IMMEDIATE, BEAT, MEAS, END				92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105,
	00 1D	0000 000a	Reset Octave Shift	(0 - 1)				106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117,
				OFF, ON				118, 119, 120, 121, 122, 123, 124, 125, 126, 127, OFF, TIE
	00 1E	0aaa aaaa	Arp Poly Remain Notes	(0 - 127)	#	00 08	0000 aaaa	
				OFF, 1, 2, 3, 4, 5, 6, 7,		00 09	0000 bbbb	[01] Value Step.04
				8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24,				(0 - 129)
				25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40,				0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11,
				41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56,				12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27,
				57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72,				28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43,
				73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88,				44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59,
				89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103,				60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75,
				104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115,				76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91,
				116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127				92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105,
								106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117,
								118, 119, 120, 121, 122, 123, 124, 125, 126, 127, OFF, TIE
	00 1F	0000 0000	Reserved	(0 - 0)				
#	00 20	0000 aaaa			#	00 0A	0000 aaaa	
	00 21	0000 bbbb				00 0B	0000 bbbb	[01] Value Step.05
	00 22	0000 cccc						(0 - 129)
	00 23	0000 dddd	Initial Pattern Number	(0 - 65535)				0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11,
				1 - 65536				12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27,
	00 24	0000 00aa	Arp Mode	(0 - 2)				28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43,
				I-ARP, ARP, STEP				44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59,
	00 25	0000 000a	Key Switch Sync	(0 - 1)				60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75,
				OFF, ON				76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91,
	00 26	0000 000a	Step Key Shift	(0 - 1)				92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105,
				OFF, ON				106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117,
	00 27	0000 0aaa	I-Arp Style Part	(0 - 4)	#	00 0C	0000 aaaa	
				PART 1, PART 2, PART 3, PART 4, PART R		00 0D	0000 bbbb	[01] Value Step.06
#	00 28	0000 aaaa						(0 - 129)
	00 29	0000 bbbb						0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11,
	00 2A	0000 cccc						12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27,
	00 2B	0000 dddd	I-Arp Style	(0 - 65)				28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43,
								44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59,
#	00 2C	0000 aaaa						60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75,
	00 2D	0000 bbbb	Arp Style	(0 - 127)				76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91,
				1 - 128				92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105,
#	00 2E	0000 aaaa						106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117,

[illegible]

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[illegible]

				106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE					106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE
#	02 34	0000	aaaa		#	02 46	0000	aaaa	
	02 35	0000	bbbb	[03] Value Step.22 (0 - 129)		02 47	0000	bbbb	[03] Value Step.31 (0 - 129)
				0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE					0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE
#	02 36	0000	aaaa		#	02 48	0000	aaaa	
	02 37	0000	bbbb	[03] Value Step.23 (0 - 129)		02 49	0000	bbbb	[03] Value Step.32 (0 - 129)
				0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE					0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE
#	02 38	0000	aaaa		#	02 4A	0000	aaaa	
	02 39	0000	bbbb	[03] Value Step.24 (0 - 129)		02 4B	0000	bbbb	[03] Value Step.33 (0 - 129)
				0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE					0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE
#	02 3A	0000	aaaa		#	02 4C	0000	aaaa	
	02 3B	0000	bbbb	[03] Value Step.25 (0 - 129)		02 4D	0000	bbbb	[03] Value Step.34 (0 - 129)
				0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE					0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,

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[illegible]

[illegible]

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				106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE					106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE	
#	04 36	0000	aaaa		#	04 48	0000	aaaa		
	04 37	0000	bbbb	[05] Value Step.19 (0 - 129)		04 49	0000	bbbb	[05] Value Step.28 (0 - 129)	
				0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE						0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE
#	04 38	0000	aaaa		#	04 4A	0000	aaaa		
	04 39	0000	bbbb	[05] Value Step.20 (0 - 129)		04 4B	0000	bbbb	[05] Value Step.29 (0 - 129)	
				0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE						0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE
#	04 3A	0000	aaaa		#	04 4C	0000	aaaa		
	04 3B	0000	bbbb	[05] Value Step.21 (0 - 129)		04 4D	0000	bbbb	[05] Value Step.30 (0 - 129)	
				0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE						0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE
#	04 3C	0000	aaaa		#	04 4E	0000	aaaa		
	04 3D	0000	bbbb	[05] Value Step.22 (0 - 129)		04 4F	0000	bbbb	[05] Value Step.31 (0 - 129)	
				0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE						0,1,2,3,4,5,6,7,8,9,10,1

				92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE					92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE		
#	04 5A 04 5B	0000 aaaa 0000 bbbb	[05] Value Step.37	(0 - 129)	#	04 6C 04 6D	0000 aaaa 0000 bbbb	[05] Value Step.46	(0 - 129)		
				0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE							0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE
#	04 5C 04 5D	0000 aaaa 0000 bbbb	[05] Value Step.38	(0 - 129)	#	04 6E 04 6F	0000 aaaa 0000 bbbb	[05] Value Step.47	(0 - 129)		
				0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE							0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE
#	04 5E 04 5F	0000 aaaa 0000 bbbb	[05] Value Step.39	(0 - 129)	#	04 70 04 71	0000 aaaa 0000 bbbb	[05] Value Step.48	(0 - 129)		
				0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE							0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE
#	04 60 04 61	0000 aaaa 0000 bbbb	[05] Value Step.40	(0 - 129)	#	04 72 04 73	0000 aaaa 0000 bbbb	[05] Value Step.49	(0 - 129)		
				0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE							0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE
#	04 62 04 63	0000 aaaa 0000 bbbb	[05] Value Step.41	(0 - 129)	#	04 74 04 75	0000 aaaa 0000 bbbb	[05] Value Step.50	(0 - 129)		
				0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE							0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE
#	04 64 04 65	0000 aaaa 0000 bbbb	[05] Value Step.42	(0 - 129)	#	04 76 04 77	0000 aaaa 0000 bbbb	[05] Value Step.51	(0 - 129)		
				0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE							0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE
#	04 66 04 67	0000 aaaa 0000 bbbb	[05] Value Step.43	(0 - 129)	#	04 78 04 79	0000 aaaa 0000 bbbb	[05] Value Step.52	(0 - 129)		
				0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE							0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE
#	04 68 04 69	0000 aaaa 0000 bbbb	[05] Value Step.44	(0 - 129)	#	04 7A 04 7B	0000 aaaa 0000 bbbb	[05] Value Step.53	(0 - 129)		
				0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE							0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE
#	04 6A 04 6B	0000 aaaa 0000 bbbb	[05] Value Step.45	(0 - 129)	#	04 7C 04 7D	0000 aaaa 0000 bbbb	[05] Value Step.54	(0 - 129)		
				0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE							0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE

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[illegible]

				76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE					76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE											
#	07 00	0000	aaaa			#	07 12	0000	aaaa											
	07 01	0000	bbbb	[07] Value Step.52	(0 - 129)		07 13	0000	bbbb	[07] Value Step.61	(0 - 129)									
					0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE															
#	07 02	0000	aaaa			#	07 14	0000	aaaa											
	07 03	0000	bbbb	[07] Value Step.53	(0 - 129)		07 15	0000	bbbb	[07] Value Step.62	(0 - 129)									
					0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE															
#	07 04	0000	aaaa			#	07 16	0000	aaaa											
	07 05	0000	bbbb	[07] Value Step.54	(0 - 129)		07 17	0000	bbbb	[07] Value Step.63	(0 - 129)									
					0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE															
#	07 06	0000	aaaa			#	07 18	0000	aaaa											
	07 07	0000	bbbb	[07] Value Step.55	(0 - 129)		07 19	0000	bbbb	[07] Value Step.64	(0 - 129)									
					0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE															
#	07 08	0000	aaaa				07 1A	0000	0000	Reserved	(0 - 0)									
	07 09	0000	bbbb	[07] Value Step.56	(0 - 129)		07 1B	0000	0000	Reserved	(0 - 0)									
					0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE	#	07 1C	0000	00aa	[08] Type										
							07 1D	0aaa	aaaa	[08] Number	(0 - 127)									
							07 1E	0000	aaaa											
							07 1F	0000	bbbb	[08] Value Step.01	(0 - 129)									
#	07 0A	0000	aaaa																	
	07 0B	0000	bbbb	[07] Value Step.57	(0 - 129)															
					0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE	#	07 20	0000	aaaa											
							07 21	0000	bbbb	[08] Value Step.02	(0 - 129)									
#	07 0C	0000	aaaa																	
	07 0D	0000	bbbb	[07] Value Step.58	(0 - 129)															
					0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE	#	07 22	0000	aaaa											
							07 23	0000	bbbb	[08] Value Step.03	(0 - 129)									
#	07 0E	0000	aaaa																	
	07 0F	0000	bbbb	[07] Value Step.59	(0 - 129)															
					0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE	#	07 24	0000	aaaa											
							07 25	0000	bbbb	[08] Value Step.04	(0 - 129)									
#	07 10	0000	aaaa																	
	07 11	0000	bbbb	[07] Value Step.60	(0 - 129)															
					0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE	#	07 26	0000	aaaa											

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[illegible]

JUNO-X MIDI Implementation

				60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE					60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE														
#	08 3A	0000	aaaa			#	08 4C	0000	aaaa			#	08 4C	0000	aaaa			#	08 4C	0000	aaaa		
	08 3B	0000	bbbb	[09] Value Step.13	(0 - 129)		08 4D	0000	bbbb	[09] Value Step.22	(0 - 129)		08 4D	0000	bbbb	[09] Value Step.22	(0 - 129)		08 4D	0000	bbbb	[09] Value Step.22	(0 - 129)
					0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE				0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE				0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE				0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE						
#	08 3C	0000	aaaa			#	08 4E	0000	aaaa			#	08 4E	0000	aaaa			#	08 4E	0000	aaaa		
	08 3D	0000	bbbb	[09] Value Step.14	(0 - 129)		08 4F	0000	bbbb	[09] Value Step.23	(0 - 129)		08 4F	0000	bbbb	[09] Value Step.23	(0 - 129)		08 4F	0000	bbbb	[09] Value Step.23	(0 - 129)
					0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE				0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE				0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52										

				60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE					60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE			
#	08 5E	0000	aaaa		#	08 70	0000	aaaa				
	08 5F	0000	bbbb	[09] Value Step.31 (0 - 129)		08 71	0000	bbbb	[09] Value Step.40 (0 - 129)			
				0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE								0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE
#	08 60	0000	aaaa		#	08 72	0000	aaaa				
	08 61	0000	bbbb	[09] Value Step.32 (0 - 129)		08 73	0000	bbbb	[09] Value Step.41 (0 - 129)			
				0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE								0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE
#	08 62	0000	aaaa		#	08 74	0000	aaaa				
	08 63	0000	bbbb	[09] Value Step.33 (0 - 129)		08 75	0000	bbbb	[09] Value Step.42 (0 - 129)			
				0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE								0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE
#	08 64	0000	aaaa		#	08 76	0000	aaaa				
	08 65	0000	bbbb	[09] Value Step.34 (0 - 129)		08 77	0000	bbbb	[09] Value Step.43 (0 - 129)			
				0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE								0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE
#	08 66	0000	aaaa		#	08 78	0000	aaaa				
	08 67	0000	bbbb	[09] Value Step.35 (0 - 129)		08 79	0000	bbbb	[09] Value Step.44 (0 - 129)			
				0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE								0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE
#	08 68	0000	aaaa		#	08 7A	0000	aaaa				
	08 69	0000	bbbb	[09] Value Step.36 (0 - 129)		08 7B	0000	bbbb	[09] Value Step.45 (0 - 129)			
				0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE								0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE
#	08 6A	0000	aaaa		#	08 7C	0000	aaaa				
	08 6B	0000	bbbb	[09] Value Step.37 (0 - 129)		08 7D	0000	bbbb	[09] Value Step.46 (0 - 129)			
				0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE								0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE
#	08 6C	0000	aaaa		#	08 7E	0000	aaaa				
	08 6D	0000	bbbb	[09] Value Step.38 (0 - 129)		08 7F	0000	bbbb	[09] Value Step.47 (0 - 129)			
				0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE								0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE
#	08 6E	0000	aaaa		#	09 00	0000	aaaa				
	08 6F	0000	bbbb	[09] Value Step.39 (0 - 129)		09 01	0000	bbbb	[09] Value Step.48 (0 - 129)			
				0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE								0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE

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#	09 28	0000	aaaa			#	09 3A	0000	aaaa		
	09 29	0000	bbbb	[10] Value Step.02	(0 - 129)		09 3B	0000	bbbb	[10] Value Step.11	(0 - 129)
					0,1,2,3,4,5,6,7,8,9,10,11,						0,1,2,3,4,5,6,7,8,9,10,11,
					12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,						12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,
					28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,						28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,
					44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59,						44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59,
					60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75,						60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75,
					76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91,						76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91,
					92,93,94,95,96,97,98,99,100,101,102,103,104,105,						92,93,94,95,96,97,98,99,100,101,102,103,104,105,
					106,107,108,109,110,111,112,113,114,115,116,117,						106,107,108,109,110,111,112,113,114,115,116,117,
					118,119,120,121,122,123,124,125,126,127,OFF,TIE						118,119,120,121,122,123,124,125,126,127,OFF,TIE
#	09 2A	0000	aaaa			#	09 3C	0000	aaaa		
	09 2B	0000	bbbb	[10] Value Step.03	(0 - 129)		09 3D	0000	bbbb	[10] Value Step.12	(0 - 129)
					0,1,2,3,4,5,6,7,8,9,10,11,						0,1,2,3,4,5,6,7,8,9,10,11,
					12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,						12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,
					28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,						28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,
					44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59,						44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59,
					60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75,						60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75,
					76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91,						76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91,
					92,93,94,95,96,97,98,99,100,101,102,103,104,105,						92,93,94,95,96,97,98,99,100,101,102,103,104,105,
					106,107,108,109,110,111,112,113,114,115,116,117,						106,107,108,109,110,111,112,113,114,115,116,117,
					118,119,120,121,122,123,124,125,126,127,OFF,TIE						118,119,120,121,122,123,124,125,126,127,OFF,TIE
#	09 2C	0000	aaaa			#	09 3E	0000	aaaa		
	09 2D	0000	bbbb	[10] Value Step.04	(0 - 129)		09 3F	0000	bbbb	[10] Value Step.13	(0 - 129)
					0,1,2,3,4,5,6,7,8,9,10,11,						0,1,2,3,4,5,6,7,8,9,10,11,
					12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,						12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,
					28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,						28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,
					44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59,						44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59,
					60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75,						60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75,
					76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91,						76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91,
					92,93,94,95,96,97,98,99,100,101,102,103,104,105,						92,93,94,95,96,97,98,99,

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#	09 70	0000	aaaa			#	0A 02	0000	aaaa		
	09 71	0000	bbbb	[10] Value Step.38	(0 - 129)		0A 03	0000	bbbb	[10] Value Step.47	(0 - 129)
				0,1,2,3,4,5,6,7,8,9,10,11,						0,1,2,3,4,5,6,7,8,9,10,11,	
				12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,						12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,	
				28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,						28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,	
				44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59,						44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59,	
				60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75,						60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75,	
				76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91,						76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91,	
				92,93,94,95,96,97,98,99,100,101,102,103,104,105,						92,93,94,95,96,97,98,99,100,101,102,103,104,105,	
				106,107,108,109,110,111,112,113,114,115,116,117,						106,107,108,109,110,111,112,113,114,115,116,117,	
				118,119,120,121,122,123,124,125,126,127,OFF,TIE						118,119,120,121,122,123,124,125,126,127,OFF,TIE	
#	09 72	0000	aaaa			#	0A 04	0000	aaaa		
	09 73	0000	bbbb	[10] Value Step.39	(0 - 129)		0A 05	0000	bbbb	[10] Value Step.48	(0 - 129)
				0,1,2,3,4,5,6,7,8,9,10,11,						0,1,2,3,4,5,6,7,8,9,10,11,	
				12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,						12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,	
				28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,						28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,	
				44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59,						44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59,	
				60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75,						60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75,	
				76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91,						76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91,	
				92,93,94,95,96,97,98,99,100,101,102,103,104,105,						92,93,94,95,96,97,98,99,100,101,102,103,104,105,	
				106,107,108,109,110,111,112,113,114,115,116,117,						106,107,108,109,110,111,112,113,114,115,116,117,	
				118,119,120,121,122,123,124,125,126,127,OFF,TIE						118,119,120,121,122,123,124,125,126,127,OFF,TIE	
#	09 74	0000	aaaa			#	0A 06	0000	aaaa		
	09 75	0000	bbbb	[10] Value Step.40	(0 - 129)		0A 07	0000	bbbb	[10] Value Step.49	(0 - 129)
				0,1,2,3,4,5,6,7,8,9,10,11,						0,1,2,3,4,5,6,7,8,9,10,11,	
				12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,						12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,	
				28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,						28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,	
				44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59,						44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59,	
				60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75,						60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75,	
				76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91,						76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91,	
				92,93,94,95,96,97,98,99,100,101,102,103,104,105,						92,93,94,95,96,97,98,99,100,101,102,103,104,105,	
				106,107,108,109,110,111,112,113,114,115,116,117,						106,107,108,109,110,111,112,113,114,115,116,117,	
				118,119,120,121,122,123,124,125,126,127,OFF,TIE						118,119,120,121,122,123,124,125,126,127,OFF,TIE	
#	09 76	0000	aaaa			#	0A 08	0000	aaaa		
	09 77	0000	bbbb	[10] Value Step.41	(0 - 129)		0A 09	0000	bbbb	[10] Value Step.50	(0 - 129)
				0,1,2,3,4,5,6,7,8,9,10,11,						0,1,2,3,4,5,6,7,8,9,10,11,	
				12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,						12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,	
				28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,						28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,	
				44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59,						44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59,	
				60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75,						60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75,	
				76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91,						76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91,	
				92,93,94,95,96,97,98,99,100,101,102,103,104,105,						92,93,94,95,96,97,98,99,100,101,102,103,104,105,	
				106,107,108,109,110,111,112,113,114,115,116,117,						106,107,108,109,110,111,112,113,114,115,116,117,	
				118,119,120,121,122,123,124,125,126,127,OFF,TIE						118,119,120,121,122,123,124,125,126,127,OFF,TIE	
#	09 78	0000	aaaa			#	0A 0A	0000	aaaa		
	09 79	0000	bbbb	[10] Value Step.42	(0 - 129)		0A 0B	0000	bbbb	[10] Value Step.51	(0 - 129)
				0,1,2,3,4,5,6,7,8,9,10,11,						0,1,2,3,4,5,6,7,8,9,10,11,	
				12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,						12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,	
				28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,						28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,	
				44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59,						44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59,	
				60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75,						60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75,	
				76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91,						76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91,	
				92,93,94,95,96,97,98,99,100,101,102,103,104,105,						92,93,94,95,96,97,98,99,100,101,102,103,104,105,	
				106,107,108,109,110,111,112,113,114,115,116,117,						106,107,108,109,110,111,112,113,114,115,116,117,	
				118,119,120,121,122,123,124,125,126,127,OFF,TIE						118,119,120,121,122,123,124,125,126,127,OFF,TIE	
#	09 7A	0000	aaaa			#	0A 0C	0000	aaaa		
	09 7B	0000	bbbb	[10] Value Step.43	(0 - 129)		0A 0D	0000	bbbb	[10] Value Step.52	(0 - 129)
				0,1,2,3,4,5,6,7,8,9,10,11,						0,1,2,3,4,5,6,7,8,9,10,11,	
				12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,						12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,	
				28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,						28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,	
				44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59,						44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59,	
				60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75,						60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75,	
				76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91,						76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91,	
				92,93,94,95,96,97,98,99,100,101,102,103,104,105,						92,93,94,95,96,97,98,99,100,101,102,103,104,105,	
				106,107,108,109,110,111,112,113,114,115,116,117,						106,107,108,109,110,111,112,113,114,115,116,117,	
				118,119,120,121,122,123,124,125,126,127,OFF,TIE						118,119,120,121,122,123,124,125,126,127,OFF,TIE	
#	09 7C	0000	aaaa			#	0A 0E	0000	aaaa		
	09 7D	0000	bbbb	[10] Value Step.44	(0 - 129)		0A 0F	0000	bbbb	[10] Value Step.53	(0 - 129)
				0,1,2,3,4,5,6,7,8,9,10,11,						0,1,2,3,4,5,6,7,8,9,10,11,	
				12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,						12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,	
				28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,						28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,	
				44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59,						44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59,	
				60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75,						60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75,	
				76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91,						76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91,	
				92,93,94,95,96,97,98,99,100,101,102,103,104,105,						92,93,94,95,96,97,98,99,100,101,102,103,104,105,	
				106,107,108,109,110,111,112,113,114,115,116,117,						106,107,108,109,110,111,112,113,114,115,116,117,	
				118,119,120,121,122,123,124,125,126,127,OFF,TIE						118,119,120,121,122,123,124,125,126,127,OFF,TIE	
#	09 7E	0000	aaaa			#	0A 10	0000	aaaa		
	09 7F	0000	bbbb	[10] Value Step.45	(0 - 129)		0A 11	0000	bbbb	[10] Value Step.54	(0 - 129)
				0,1,2,3,4,5,6,7,8,9,10,11,						0,1,2,3,4,5,6,7,8,9,10,11,	
				12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,						12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,	
				28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,						28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,	
				44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59,						44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59,	
				60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75,						60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75,	
				76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91,						76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91,	
				92,93,94,95,96,97,98,99,100,101,102,103,104,105,						92,93,94,95,96,97,98,99,100,101,102,103,104,105,	
				106,107,108,109,110,111,112,113,114,115,116,117,						106,107,108,109,110,111,112,113,114,115,116,117,	
				118,119,120,121,122,123,124,125,126,127,OFF,TIE						118,119,120,121,122,123,124,125,126,127,OFF,TIE	
#	0A 00	0000	aaaa			#	0A 12	0000	aaaa		
	0A 01	0000	bbbb	[10] Value Step.46	(0 - 129)		0A 13	0000	bbbb	[10] Value Step.55	(0 - 129)
				0,1,2,3,4,5,6,7,8,9,10,11,						0,1,2,3,4,5,6,7,8,9,10,11,	
				12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,						12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,	
				28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,						28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,	
				44,45,46,47,							

				44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, OFF, TIE								44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, OFF, TIE															
#	OA 3C OA 3D	0000 aaaa 0000 bbbb	[11]	Value Step.10	(0 - 129)				#	OA 4E OA 4F	0000 aaaa 0000 bbbb	[11]	Value Step.19	(0 - 129)													
					0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11,									0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11,													
					12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27,									12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27,													
					28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43,									28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43,													
					44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59,									44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59,													
					60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75,									60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75,													
					76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91,									76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91,													
					92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105,									92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105,													
					106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117,									106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117,													
					118, 119, 120, 121, 122, 123, 124, 125, 126, 127, OFF, TIE									118, 119, 120, 121, 122, 123, 124, 125, 126, 127, OFF, TIE													
#	OA 3E OA 3F	0000 aaaa 0000 bbbb	[11]	Value Step.11	(0 - 129)				#	OA 50 OA 51	0000 aaaa 0000 bbbb	[11]	Value Step.20	(0 - 129)													
					0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11,									0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11,													
					12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27,									12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27,													
					28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43,									28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43,													
					44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59,									44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59,													
					60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75,									60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75,													
					76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91,									76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91,													
					92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105,									92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105,													
					106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117,									106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117,													
					118, 119, 120, 121, 122, 123, 124, 125, 126, 127, OFF, TIE									118, 119, 120, 121, 122, 123, 124, 125, 126, 127, OFF, TIE													
#	OA 40 OA 41	0000 aaaa 0000 bbbb	[11]	Value Step.12	(0 - 129)				#	OA 52 OA 53	0000 aaaa 0000 bbbb	[11]	Value Step.21	(0 - 129)													
					0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11,									0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11,													
					12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27,									12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27,													
					28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43,									28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43,													
					44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59,									44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59,													
					60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75,									60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75,													
					76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91,									76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91,													
					92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105,									92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105,													
					106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117,									106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117,													
					118, 119, 120, 121, 122, 123, 124, 125, 126, 127, OFF, TIE									118, 119, 120, 121, 122, 123, 124, 125, 126, 127, OFF, TIE													
#	OA 42 OA 43	0000 aaaa 0000 bbbb	[11]	Value Step.13	(0 - 129)				#	OA 54 OA 55	0000 aaaa 0000 bbbb	[11]	Value Step.22	(0 - 129)													
					0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11,									0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11,													
					12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27,									12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27,													
					28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43,									28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43,													
					44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59,									44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59,													
					60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75,									60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75,													
					76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91,									76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91,													
					92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105,									92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105,													
					106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117,									106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117,													
					118, 119, 120, 121, 122, 123, 124, 125, 126, 127, OFF, TIE									118, 119, 120, 121, 122, 123, 124, 125, 126, 127, OFF, TIE													
#	OA 44 OA 45	0000 aaaa 0000 bbbb	[11]	Value Step.14	(0 - 129)				#	OA 56 OA 57	0000 aaaa 0000 bbbb	[11]	Value Step.23	(0 - 129)													
					0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11,									0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11,													
					12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27,									12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27,													
					28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43,									28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43,													
					44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59,									44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59,													
					60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75,									60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75,													
					76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91,									76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91,													
					92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105,									92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105,													
					106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117,									106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117,													
					118, 119, 120, 121, 122, 123, 124, 125, 126, 127, OFF, TIE									118, 119, 120, 121, 122, 123, 124, 125, 126, 127, OFF, TIE													
#	OA 46 OA 47	0000 aaaa 0000 bbbb	[11]	Value Step.15	(0 - 129)				#	OA 58 OA 59	0000 aaaa 0000 bbbb	[11]	Value Step.24	(0 - 129)													
		</																									

[illegible]

#	OB 4E	0000 aaaa	118,119,120,121,122,123,124,125,126,127,OFF,TIE	#	OB 60	0000 aaaa	118,119,120,121,122,123,124,125,126,127,OFF,TIE
	OB 4F	0000 bbbb	[12] Value Step.17 (0 - 129)		OB 61	0000 bbbb	[12] Value Step.26 (0 - 129)
			0,1,2,3,4,5,6,7,8,9,10,11,				0,1,2,3,4,5,6,7,8,9,10,11,
			12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,				12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,
			28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,				28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,
			44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59,				44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59,
			60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75,				60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75,
			76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91,				76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91,
			92,93,94,95,96,97,98,99,100,101,102,103,104,105,				92,93,94,95,96,97,98,99,100,101,102,103,104,105,
			106,107,108,109,110,111,112,113,114,115,116,117,				106,107,108,109,110,111,112,113,114,115,116,117,
			118,119,120,121,122,123,124,125,126,127,OFF,TIE				118,119,120,121,122,123,124,125,126,127,OFF,TIE
#	OB 50	0000 aaaa		#	OB 62	0000 aaaa	
	OB 51	0000 bbbb	[12] Value Step.18 (0 - 129)		OB 63	0000 bbbb	[12] Value Step.27 (0 - 129)
			0,1,2,3,4,5,6,7,8,9,10,11,				0,1,2,3,4,5,6,7,8,9,10,11,
			12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,				12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,
			28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,				28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,
			44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59,				44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59,
			60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75,				60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75,
			76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91,				76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91,
			92,93,94,95,96,97,98,99,100,101,102,103,104,105,				92,93,94,95,96,97,98,99,100,101,102,103,104,105,
			106,107,108,109,110,111,112,113,114,115,116,117,				106,107,108,109,110,111,112,113,114,115,116,117,
			118,119,120,121,122,123,124,125,126,127,OFF,TIE				118,119,120,121,122,123,124,125,126,127,OFF,TIE
#	OB 52	0000 aaaa		#	OB 64	0000 aaaa	
	OB 53	0000 bbbb	[12] Value Step.19 (0 - 129)		OB 65	0000 bbbb	[12] Value Step.28 (0 - 129)
			0,1,2,3,4,5,6,7,8,9,10,11,				0,1,2,3,4,5,6,7,8,9,10,11,
			12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,				12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,
			28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,				28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,
			44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59,				44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59,
			60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75,				60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75,
			76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91,				76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91,
			92,93,94,95,96,97,98,99,100,101,102,103,104,105,				92,93,94,95,96,97,98,99,100,101,102,103,104,105,
			106,107,108,109,110,111,112,113,114,115,116,117,				106,107,108,109,110,111,112,113,114,115,116,117,
			118,119,120,121,122,123,124,125,126,127,OFF,TIE				118,119,120,121,122,123,124,125,126,127,OFF,TIE
#	OB 54	0000 aaaa		#	OB 66	0000 aaaa	
	OB 55	0000 bbbb	[12] Value Step.20 (0 - 129)		OB 67	0000 bbbb	[12

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#				118, 119, 120, 121, 122, 123, 124, 125, 126, 127, OFF, TIE	#				118, 119, 120, 121, 122, 123, 124, 125, 126, 127, OFF, TIE	
	OB 72	0000	aaaa				OC 04	0000	aaaa	
	OB 73	0000	bbbb	[12] Value Step.35 (0 - 129)			OC 05	0000	bbbb	[12] Value Step.44 (0 - 129)
				0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11,						0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11,
				12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27,						12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27,
				28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43,						28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43,
				44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59,						44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59,
				60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75,						60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75,
				76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91,						76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91,
				92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105,						92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105,
				106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117,						106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117,
				118, 119, 120, 121, 122, 123, 124, 125, 126, 127, OFF, TIE						118, 119, 120, 121, 122, 123, 124, 125, 126, 127, OFF, TIE
#	OB 74	0000	aaaa		#	OC 06	0000	aaaa		
	OB 75	0000	bbbb	[12] Value Step.36 (0 - 129)		OC 07	0000	bbbb	[12] Value Step.45 (0 - 129)	
				0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11,						0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11,
				12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27,						12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27,
				28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43,						28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43,
				44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59,						44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59,
				60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75,						60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75,
				76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91,						76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91,
				92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105,						92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105,
				106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117,						106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117,
				118, 119, 120, 121, 122, 123, 124, 125, 126, 127, OFF, TIE						118, 119, 120, 121, 122, 123, 124, 125, 126, 127, OFF, TIE
#	OB 76	0000	aaaa		#	OC 08	0000	aaaa		
	OB 77	0000	bbbb	[12] Value Step.37 (0 - 129)		OC 09	0000	bbbb	[12] Value Step.46 (0 - 129)	
				0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11,						0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11,
				12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27,						12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27,
				28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43,						28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43,
				44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59,						44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59,
				60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75,						60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75,
				76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91,						76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91,
				92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105,						92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105,
				106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117,						106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117,
				118, 119, 120, 121, 122, 123, 124, 125, 126, 127, OFF, TIE						118, 119, 120, 121, 122, 123, 124, 125, 126, 127, OFF, TIE
#	OB 78	0000	aaaa		#	OC 0A	0000	aaaa		
	OB 79	0000	bbbb	[12] Value Step.38 (0 - 129)		OC 0B	0000	bbbb	[12] Value Step.47 (0 - 129)	
				0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11,						0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11,
				12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27,						12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27,
				28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43,						28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43,
				44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59,						44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59,
				60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75,						60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75,
				76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91,						76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91,
				92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105,						92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105,
				106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117,						106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117,
				118, 119, 120, 121, 122, 123, 124, 125, 126, 127, OFF, TIE						118, 119, 120, 121, 122, 123, 124, 125, 126, 127, OFF, TIE
#	OB 7A	0000	aaaa		#	OC 0C	0000	aaaa		
	OB 7B	0000	bbbb	[12] Value Step.39 (0 - 129)		OC 0D	0000	bbbb	[12] Value Step.48 (0 - 129)	
				0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11,						0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11,
				12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27,						12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27,
				28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43,						28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43,
				44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59,						44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59,
				60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75,						60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75,
				76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91,						76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91,
				92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105,						92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105,
				106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117,						106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117,
				118, 119, 120, 121, 122, 123, 124, 125, 126, 127, OFF, TIE						118, 119, 120, 121, 122, 123, 124, 125, 126, 127, OFF, TIE
#	OB 7C	0000	aaaa		#	OC 0E	0000	aaaa		
	OB 7D	0000	bbbb	[12] Value Step.40 (0 - 129)		OC 0F	0000	bbbb	[12] Value Step.49 (0 - 129)	
				0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11,						0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11,
				12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27,						12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27,
				28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43,						28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43,
				44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59,						44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59,
				60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75,						60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75,
				76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91,						76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91,
				92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105,						92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105,
				106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117,						106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117,
				118, 119, 120, 121, 122, 123, 124, 125, 126, 127, OFF, TIE						118, 119, 120, 121, 122, 123, 124, 125, 126, 127, OFF, TIE
#	OB 7E	0000	aaaa		#	OC 10	0000	aaaa		
	OB 7F	0000	bbbb	[12] Value Step.41 (0 - 129)		OC 11	0000	bbbb	[12] Value Step.50 (0 - 129)	
				0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11,						0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11,
				12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27,						12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27,
				28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43,						28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43,
				44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59,						44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59,
				60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75,						60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75,
				76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91,						76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91,
				92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105,						92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105,
				106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117,						106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117,
				118, 119, 120, 121, 122, 123, 124, 125, 126, 127, OFF, TIE						118, 119, 120, 121, 122, 123, 124, 125, 126, 127, OFF, TIE
#	OC 00	0000	aaaa		#	OC 12	0000	aaaa		
	OC 01	0000	bbbb	[12] Value Step.42 (0 - 129)		OC 13	0000	bbbb	[12] Value Step.51 (0 - 129)	
				0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11,						0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11,
				12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27,						12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27,
				28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43,						28, 29, 30, 31,

61

[illegible]

64

[illegible]

66

				106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE					106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE	
#	OD 74	0000	aaaa		#	OE 06	0000	aaaa		
	OD 75	0000	bbbb	[14] Value Step.32 (0 - 129)		OE 07	0000	bbbb	[14] Value Step.41 (0 - 129)	
				0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE						0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE
#	OD 76	0000	aaaa		#	OE 08	0000	aaaa		
	OD 77	0000	bbbb	[14] Value Step.33 (0 - 129)		OE 09	0000	bbbb	[14] Value Step.42 (0 - 129)	
				0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE						0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE
#	OD 78	0000	aaaa		#	OE 0A	0000	aaaa		
	OD 79	0000	bbbb	[14] Value Step.34 (0 - 129)		OE 0B	0000	bbbb	[14] Value Step.43 (0 - 129)	
				0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE						0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE
#	OD 7A	0000	aaaa		#	OE 0C	0000	aaaa		
	OD 7B	0000	bbbb	[14] Value Step.35 (0 - 129)		OE 0D	0000	bbbb	[14] Value Step.44 (0 - 129)	
				0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE						0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE
#	OD 7C	0000	aaaa		#	OE 0E	0000	aaaa		
	OD 7D	0000	bbbb	[14] Value Step.36 (0 - 129)		OE 0F	0000	bbbb	[14] Value Step.45 (0 - 129)	
				0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE						0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE
#	OD 7E	0000	aaaa		#	OE 10	0000	aaaa		
	OD 7F	0000	bbbb	[14] Value Step.37 (0 - 129)		OE 11	0000	bbbb	[14] Value Step.46 (0 - 129)	
				0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE						0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE
#	OE 00	0000	aaaa		#	OE 12	0000	aaaa		
	OE 01	0000	bbbb	[14] Value Step.38 (0 - 129)		OE 13	0000	bbbb	[14] Value Step.47 (0 - 129)	
				0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE						0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE
#	OE 02	0000	aaaa		#	OE 14	0000	aaaa		
	OE 03	0000	bbbb	[14] Value Step.39 (0 - 129)		OE 15	0000	bbbb	[14] Value Step.48 (0 - 129)	
				0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE						0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE
#	OE 04	0000	aaaa		#	OE 16	0000	aaaa		
	OE 05	0000	bbbb	[14] Value Step.40 (0 - 129)		OE 17	0000	bbbb	[14] Value Step.49 (0 - 129)	
				0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE						0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE

[illegible]

70

[illegible]

JUNO-X MIDI Implementation

[illegible]

				92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE					92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE	
#	0F 52	0000	aaaa		#	0F 64	0000	aaaa		
	0F 53	0000	bbbb	[16] Value Step.11 (0 - 129)		0F 65	0000	bbbb	[16] Value Step.20 (0 - 129)	
				0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE						0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE
#	0F 54	0000	aaaa		#	0F 66	0000	aaaa		
	0F 55	0000	bbbb	[16] Value Step.12 (0 - 129)		0F 67	0000	bbbb	[16] Value Step.21 (0 - 129)	
				0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE						0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE
#	0F 56	0000	aaaa		#	0F 68	0000	aaaa		
	0F 57	0000	bbbb	[16] Value Step.13 (0 - 129)		0F 69	0000	bbbb	[16] Value Step.22 (0 - 129)	
				0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE						0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE
#	0F 58	0000	aaaa		#	0F 6A	0000	aaaa		
	0F 59	0000	bbbb	[16] Value Step.14 (0 - 129)		0F 6B	0000	bbbb	[16] Value Step.23 (0 - 129)	
				0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE						0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE
#	0F 5A	0000	aaaa		#	0F 6C	0000	aaaa		
	0F 5B	0000	bbbb	[16] Value Step.15 (0 - 129)		0F 6D	0000	bbbb	[16] Value Step.24 (0 - 129)	
				0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE						0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE
#	0F 5C	0000	aaaa		#	0F 6E	0000	aaaa		
	0F 5D	0000	bbbb	[16] Value Step.16 (0 - 129)		0F 6F	0000	bbbb	[16] Value Step.25 (0 - 129)	
				0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE						0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE
#	0F 5E	0000	aaaa		#	0F 70	0000	aaaa		
	0F 5F	0000	bbbb	[16] Value Step.17 (0 - 129)		0F 71	0000	bbbb	[16] Value Step.26 (0 - 129)	
				0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE						0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE
#	0F 60	0000	aaaa		#	0F 72	0000	aaaa		
	0F 61	0000	bbbb	[16] Value Step.18 (0 - 129)		0F 73	0000	bbbb	[16] Value Step.27 (0 - 129)	
				0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE						0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE
#	0F 62	0000	aaaa		#	0F 74	0000	aaaa		
	0F 63	0000	bbbb	[16] Value Step.19 (0 - 129)		0F 75	0000	bbbb	[16] Value Step.28 (0 - 129)	
				0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE						0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE

			106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE				106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE	
#	0F 76	0000	aaaa		#	10 08	0000	aaaa
	0F 77	0000	bbbb	[16] Value Step.29		10 09	0000	bbbb
				(0 - 129)				[16] Value Step.38
				0,1,2,3,4,5,6,7,8,9,10,11,				(0 - 129)
				12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,				0,1,2,3,4,5,6,7,8,9,10,11,
				28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,				12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,
				44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59,				28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,
				60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75,				44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59,
				76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91,				60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75,
				92,93,94,95,96,97,98,99,100,101,102,103,104,105,				76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91,
				106,107,108,109,110,111,112,113,114,115,116,117,				92,93,94,95,96,97,98,99,100,101,102,103,104,105,
				118,119,120,121,122,123,124,125,126,127,OFF,TIE				106,107,108,109,110,111,112,113,114,115,116,117,
#	0F 78	0000	aaaa		#	10 0A	0000	aaaa
	0F 79	0000	bbbb	[16] Value Step.30		10 0B	0000	bbbb
				(0 - 129)				[16] Value Step.39
				0,1,2,3,4,5,6,7,8,9,10,11,				(0 - 129)
				12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,				0,1,2,3,4,5,6,7,8,9,10,11,
				28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,				12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,
				44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59,				28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,
				60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75,				44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59,
				76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91,				60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75,
				92,93,94,95,96,97,98,99,100,101,102,103,104,105,				76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91,
				106,107,108,109,110,111,112,113,114,115,116,117,				92,93,94,95,96,97,98,99,100,101,102,103,104,105,
				118,119,120,121,122,123,124,125,126,127,OFF,TIE				106,107,108,109,110,111,112,113,114,115,116,117,
#	0F 7A	0000	aaaa		#	10 0C	0000	aaaa
	0F 7B	0000	bbbb	[16] Value Step.31		10 0D	0000	bbbb
				(0 - 129)				[16] Value Step.40
				0,1,2,3,4,5,6,7,8,9,10,11,				(0 - 129)
				12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,				0,1,2,3,4,5,6,7,8,9,10,11,
				28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,				12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,
				44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59,				28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,
				60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75,				44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59,
				76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91,				60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75,
				92,93,94,95,96,97,98,99,100,101,102,103,104,105,				76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91,
				106,107,108,109,110,111,112,113,114,115,116,117,				92,93,94,95,96,97,98,99,100,101,102,103,104,105,

				92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE					92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE	
#	10 1A	0000	aaaa		#	10 2C	0000	aaaa		
	10 1B	0000	bbbb	[16] Value Step.47 (0 - 129)		10 2D	0000	bbbb	[16] Value Step.56 (0 - 129)	
				0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE						0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE
#	10 1C	0000	aaaa		#	10 2E	0000	aaaa		
	10 1D	0000	bbbb	[16] Value Step.48 (0 - 129)		10 2F	0000	bbbb	[16] Value Step.57 (0 - 129)	
				0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE						0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE
#	10 1E	0000	aaaa		#	10 30	0000	aaaa		
	10 1F	0000	bbbb	[16] Value Step.49 (0 - 129)		10 31	0000	bbbb	[16] Value Step.58 (0 - 129)	
				0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE						0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE
#	10 20	0000	aaaa		#	10 32	0000	aaaa		
	10 21	0000	bbbb	[16] Value Step.50 (0 - 129)		10 33	0000	bbbb	[16] Value Step.59 (0 - 129)	
				0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE						0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE
#	10 22	0000	aaaa		#	10 34	0000	aaaa		
	10 23	0000	bbbb	[16] Value Step.51 (0 - 129)		10 35	0000	bbbb	[16] Value Step.60 (0 - 129)	
				0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE						0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE
#	10 24	0000	aaaa		#	10 36	0000	aaaa		
	10 25	0000	bbbb	[16] Value Step.52 (0 - 129)		10 37	0000	bbbb	[16] Value Step.61 (0 - 129)	
				0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE						0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE
#	10 26	0000	aaaa		#	10 38	0000	aaaa		
	10 27	0000	bbbb	[16] Value Step.53 (0 - 129)		10 39	0000	bbbb	[16] Value Step.62 (0 - 129)	
				0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE						0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE
#	10 28	0000	aaaa		#	10 3A	0000	aaaa		
	10 29	0000	bbbb	[16] Value Step.54 (0 - 129)		10 3B	0000	bbbb	[16] Value Step.63 (0 - 129)	
				0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE						0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE
#	10 2A	0000	aaaa		#	10 3C	0000	aaaa		
	10 2B	0000	bbbb	[16] Value Step.55 (0 - 129)		10 3D	0000	bbbb	[16] Value Step.64 (0 - 129)	
				0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE						0,1,2,3,4,5,6,7,8,9,10,11, 12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75, 76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91, 92,93,94,95,96,97,98,99,100,101,102,103,104,105, 106,107,108,109,110,111,112,113,114,115,116,117, 118,119,120,121,122,123,124,125,126,127,OFF,TIE

JUNO-X MIDI Implementation

			92,93,94,95,96,97,98,99,100,101,102,103,104,105,
			106,107,108,109,110,111,112,113,114,115,116,117,
			118,119,120,121,122,123,124,125,126,127,OFF,TIE
10 3E	0000 0000	Reserved	(0 - 0)
10 3F	0000 0000	Reserved	(0 - 0)

00 00 10 40	Total Size		

* [Master Comp]

Offset	Address	Description	
00 00	0000 000a	Switch	(0 - 1) OFF, ON
00 01	0aaa aaaa	Low Attack Time	(0 - 100) 0.1,1,2,3,4, 5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22, 23,24,25,26,27,28,29,30,31,32,33,34,35,36,37,38, 39,40,41,42,43,44,45,46,47,48,49,50,51,52,53,54, 55,56,57,58,59,60,61,62,63,64,65,66,67,68,69,70, 71,72,73,74,75,76,77,78,79,80,81,82,83,84,85,86, 87,88,89,90,91,92,93,94,95,96,97,98,99,100 [ms]
00 02	0aaa aaaa	Low Release Time	(0 - 99) 10 - 1000 [ms]
00 03	0aaa aaaa	Low Threshold	(4 - 64) -60 - 0 [dB]
00 04	0000 0aaa	Low Ratio	(0 - 7) 1:1, 2:1, 3:1, 4:1, 8:1, 16:1, 32:1, INF:1
00 05	000a aaaa	Low Knee	(0 - 30) 0 - 30 [dB]
00 06	0aaa aaaa	Low Output Gain	(16 - 112) -24.0 - +24.0 [dB]
00 07	0aaa aaaa	Mid Attack Time	(0 - 100) 0.1,1,2,3,4, 5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22, 23,24,25,26,27,28,29,30,31,32,33,34,35,36,37,38, 39,40,41,42,43,44,45,46,47,48,49,50,51,52,53,54, 55,56,57,58,59,60,61,62,63,64,65,66,67,68,69,70, 71,72,73,74,75,76,77,78,79,80,81,82,83,84,85,86, 87,88,89,90,91,92,93,94,95,96,97,98,99,100 [ms]
00 08	0aaa aaaa	Mid Release Time	(0 - 99) 10 - 1000 [ms]
00 09	0aaa aaaa	Mid Threshold	(4 - 64) -60 - 0 [dB]
00 0A	0000 0aaa	Mid Ratio	(0 - 7) 1:1, 2:1, 3:1, 4:1, 8:1, 16:1, 32:1, INF:1
00 0B	000a aaaa	Mid Knee	(0 - 30) 0 - 30 [dB]
00 0C	0aaa aaaa	Mid Output Gain	(16 - 112) -24.0 - +24.0 [dB]
00 0D	0aaa aaaa	High Attack Time	(0 - 100) 0.1,1,2,3,4, 5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22, 23,24,25,26,27,28,29,30,31,32,33,34,35,36,37,38, 39,40,41,42,43,44,45,46,47,48,49,50,51,52,53,54, 55,56,57,58,59,60,61,62,63,64,65,66,67,68,69,70, 71,72,73,74,75,76,77,78,79,80,81,82,83,84,85,86, 87,88,89,90,91,92,93,94,95,96,97,98,99,100 [ms]
00 0E	0aaa aaaa	High Release Time	(0 - 99) 10 - 1000 [ms]
00 0F	0aaa aaaa	High Threshold	(4 - 64) -60 - 0 [dB]
00 10	0000 0aaa	High Ratio	(0 - 7) 1:1, 2:1, 3:1, 4:1, 8:1, 16:1, 32:1, INF:1
00 11	000a aaaa	High Knee	(0 - 30) 0 - 30 [dB]
00 12	0aaa aaaa	High Output Gain	(16 - 112) -24.0 - +24.0 [dB]
00 13	000a aaaa	Split Freq Low	(0 - 30) 16,20,25,31,40,50,63,80,100,125,160,200, 250,315,400,500,630,800,1000,1250,1600,2000,2500, 3150,4000,5000,6300,8000,10000,12500,16000 [Hz]
00 14	000a aaaa	Split Freq Hi	(0 - 30) 16,20,25,31,40,50,63,80,100,125,160,200, 250,315,400,500,630,800,1000,1250,1600,2000,2500, 3150,4000,5000,6300,8000,10000,12500,16000 [Hz]

00 00 00 15	Total Size		

* [Master EQ]

Offset	Address	Description	
00 00	0aaa aaaa	EQ Input Gain	(40 - 88) -24 - +24 [dB]
00 01	0aaa aaaa	EQ Low Gain	(40 - 88) -24 - +24 [dB]
00 02	0aaa aaaa	EQ Mid1 Gain	(40 - 88) -24 - +24 [dB]
00 03	0aaa aaaa	EQ Mid2 Gain	(40 - 88) -24 - +24 [dB]
00 04	0aaa aaaa	EQ Mid3 Gain	(40 - 88) -24 - +24 [dB]
00 05	0aaa aaaa	EQ High Gain	(40 - 88) -24 - +24 [dB]

00 06	000a aaaa	EQ Low Frequency	(0 - 29)
		20,25,31,40,50,63,80,100,125,160,200,	
		250,315,400,500,630,800,1000,1250,1600,2000,2500,	
		3150,4000,5000,6300,8000,10000,12600,16000 [Hz]	
00 07	000a aaaa	EQ Mid1 Frequency	(0 - 29)
		20,25,31,40,50,63,80,100,125,160,200,	
		250,315,400,500,630,800,1000,1250,1600,2000,2500,	
		3150,4000,5000,6300,8000,10000,12600,16000 [Hz]	
00 08	000a aaaa	EQ Mid2 Frequency	(0 - 29)
		20,25,31,40,50,63,80,100,125,160,200,	
		250,315,400,500,630,800,1000,1250,1600,2000,2500,	
		3150,4000,5000,6300,8000,10000,12600,16000 [Hz]	
00 09	000a aaaa	EQ Mid3 Frequency	(0 - 29)
		20,25,31,40,50,63,80,100,125,160,200,	
		250,315,400,500,630,800,1000,1250,1600,2000,2500,	
		3150,4000,5000,6300,8000,10000,12600,16000 [Hz]	
00 0A	000a aaaa	EQ High Frequency	(0 - 29)
		20,25,31,40,50,63,80,100,125,160,200,	
		250,315,400,500,630,800,1000,1250,1600,2000,2500,	
		3150,4000,5000,6300,8000,10000,12600,16000 [Hz]	
00 0B	0000 0aaa	EQ Mid1 Q	(0 - 5)
		0.5, 1.0, 2.0, 4.0, 8.0, 16.0	
00 0C	0000 0aaa	EQ Mid2 Q	(0 - 5)
		0.5, 1.0, 2.0, 4.0, 8.0, 16.0	
00 0D	0000 0aaa	EQ Mid3 Q	(0 - 5)
		0.5, 1.0, 2.0, 4.0, 8.0, 16.0	
00 0E	0000 000a	EQ Switch	(0 - 1)
		OFF, ON	

00 00 00 0F	Total Size		

* [Sympathetic Resonance]

Offset	Address	Description	
00 00	0000 000a	SymReso Switch	(0 - 1) OFF, ON
00 01	0000 0000	Reserved	(0 - 0)
00 02	0aaa aaaa	SymReso Depth	(0 - 127)
00 03	0aaa aaaa	Cabinet Resonance	(0 - 127)

00 00 00 04	Total Size		

* [ToneName]

Offset			Address		Description		
	00 00	0aaa aaaa	Name			(32 - 127)	
						32 - 127 [ASCII]	
	00 01	0aaa aaaa	Name			(32 - 127)	
						32 - 127 [ASCII]	
	00 02	0aaa aaaa	Name			(32 - 127)	
						32 - 127 [ASCII]	
	00 03	0aaa aaaa	Name			(32 - 127)	
						32 - 127 [ASCII]	
	00 04	0aaa aaaa	Name			(32 - 127)	
						32 - 127 [ASCII]	
	00 05	0aaa aaaa	Name			(32 - 127)	
						32 - 127 [ASCII]	
	00 06	0aaa aaaa	Name			(32 - 127)	
						32 - 127 [ASCII]	
	00 07	0aaa aaaa	Name			(32 - 127)	
						32 - 127 [ASCII]	
	00 08	0aaa aaaa	Name			(32 - 127)	
						32 - 127 [ASCII]	
	00 09	0aaa aaaa	Name			(32 - 127)	
						32 - 127 [ASCII]	
	00 0A	0aaa aaaa	Name			(32 - 127)	
						32 - 127 [ASCII]	
	00 0B	0aaa aaaa	Name			(32 - 127)	
						32 - 127 [ASCII]	
	00 0C	0aaa aaaa	Name			(32 - 127)	
						32 - 127 [ASCII]	
	00 0D	0aaa aaaa	Name			(32 - 127)	
						32 - 127 [ASCII]	
	00 0E	0aaa aaaa	Name			(32 - 127)	
						32 - 127 [ASCII]	
	00 0F	0aaa aaaa	Name			(32 - 127)	
						32 - 127 [ASCII]	

00 00 00 10	Total Size						

* [ToneCom]

Offset	Address	Description	
00 00	0aaa aaaa	Name	(32 - 127) 32 - 127 [ASCII]
00 01	0aaa aaaa	Name	(32 - 127) 32 - 127 [ASCII]
00 02	0aaa aaaa	Name	(32 - 127) 32 - 127 [ASCII]
00 03	0aaa aaaa	Name	(32 - 127) 32 - 127 [ASCII]

			32 - 127 [ASCII]		00 18 0000 dddd EXZ	(0 - 65535)
	00 04	0aaa aaaa	Name (32 - 127)	#	00 19 0000 aaaa	
			32 - 127 [ASCII]		00 1A 0000 bbbb	
	00 05	0aaa aaaa	Name (32 - 127)		00 1B 0000 cccc	
			32 - 127 [ASCII]		00 1C 0000 dddd EXZ	(0 - 65535)
	00 06	0aaa aaaa	Name (32 - 127)	#	00 1D 0000 aaaa	
			32 - 127 [ASCII]		00 1E 0000 bbbb	
	00 07	0aaa aaaa	Name (32 - 127)		00 1F 0000 cccc	
			32 - 127 [ASCII]		00 20 0000 dddd EXZ	(0 - 65535)
	00 08	0aaa aaaa	Name (32 - 127)	#	00 21 0000 aaaa	
			32 - 127 [ASCII]		00 22 0000 bbbb	
	00 09	0aaa aaaa	Name (32 - 127)		00 23 0000 cccc	
			32 - 127 [ASCII]		00 24 0000 dddd EXZ	(0 - 65535)
	00 0A	0aaa aaaa	Name (32 - 127)	#	00 25 0000 aaaa	
			32 - 127 [ASCII]		00 26 0000 bbbb	
	00 0B	0aaa aaaa	Name (32 - 127)		00 27 0000 cccc	
			32 - 127 [ASCII]		00 28 0000 dddd EXZ	(0 - 65535)
	00 0C	0aaa aaaa	Name (32 - 127)	#	00 29 0000 aaaa	
			32 - 127 [ASCII]		00 2A 0000 bbbb	
	00 0D	0aaa aaaa	Name (32 - 127)		00 2B 0000 cccc	
			32 - 127 [ASCII]		00 2C 0000 dddd EXZ	(0 - 65535)
	00 0E	0aaa aaaa	Name (32 - 127)	#	00 2D 0000 aaaa	
			32 - 127 [ASCII]		00 2E 0000 bbbb	
	00 0F	0aaa aaaa	Name (32 - 127)		00 2F 0000 cccc	
			32 - 127 [ASCII]		00 30 0000 dddd EXZ	(0 - 65535)
	00 10	00aa aaaa	Category (0 - 50)		-----	
			0 - 50		00 00 00 31	Total Size
#	00 11	0000 aaaa			-----	
	00 12	0000 bbbb	Reserved (0 - 0)		-----	
#	00 13	0000 aaaa			-----	
	00 14	0000 bbbb			-----	
	00 15	0000 cccc			-----	
	00 16	0000 dddd	Reserved (0 - 0)		-----	
	00 00 00 17	Total Size			-----	

* [Model Bank Assign]						

	Offset					
	Address		Description			

	00 00	0000 00aa	Attribute (0 - 3)			
			MODEL, CATEGORY, USER, EXZ			
	00 01	0000 aaaa	Model (0 - 13)			
			OFF, NO ASSIGN, COMMON,			
			PR-A, PR-B, PR-C, PR-D, XV-5080, JUNO-X, JUNO-106,			
			JUNO-60, VOCODER, RD-PIANO, PR-X			
	00 02	0000 aaaa	Model (0 - 13)			
			OFF, NO ASSIGN, COMMON,			
			PR-A, PR-B, PR-C, PR-D, XV-5080, JUNO-X, JUNO-106,			
			JUNO-60, VOCODER, RD-PIANO, PR-X			
	00 03	0000 aaaa	Model (0 - 13)			
			OFF, NO ASSIGN, COMMON,			
			PR-A, PR-B, PR-C, PR-D, XV-5080, JUNO-X, JUNO-106,			
			JUNO-60, VOCODER, RD-PIANO, PR-X			
	00 04	0000 aaaa	Model (0 - 13)			
			OFF, NO ASSIGN, COMMON,			
			PR-A, PR-B, PR-C, PR-D, XV-5080, JUNO-X, JUNO-106,			
			JUNO-60, VOCODER, RD-PIANO, PR-X			
	00 05	0000 aaaa	Model (0 - 13)			
			OFF, NO ASSIGN, COMMON,			
			PR-A, PR-B, PR-C, PR-D, XV-5080, JUNO-X, JUNO-106,			
			JUNO-60, VOCODER, RD-PIANO, PR-X			
	00 06	0000 aaaa	Model (0 - 13)			
			OFF, NO ASSIGN, COMMON,			
			PR-A, PR-B, PR-C, PR-D, XV-5080, JUNO-X, JUNO-106,			
			JUNO-60, VOCODER, RD-PIANO, PR-X			
	00 07	0000 aaaa	Model (0 - 13)			
			OFF, NO ASSIGN, COMMON,			
			PR-A, PR-B, PR-C, PR-D, XV-5080, JUNO-X, JUNO-106,			
			JUNO-60, VOCODER, RD-PIANO, PR-X			
	00 08	0000 aaaa	Model (0 - 13)			
			OFF, NO ASSIGN, COMMON,			
			PR-A, PR-B, PR-C, PR-D, XV-5080, JUNO-X, JUNO-106,			
			JUNO-60, VOCODER, RD-PIANO, PR-X			
	00 09	00aa aaaa	Category (0 - 50)			
			0 - 50			
	00 0A	00aa aaaa	Category (0 - 50)			
			0 - 50			
	00 0B	00aa aaaa	Category (0 - 50)			
			0 - 50			
	00 0C	00aa aaaa	Category (0 - 50)			
			0 - 50			
	00 0D	00aa aaaa	Category (0 - 50)			
			0 - 50			
	00 0E	00aa aaaa	Category (0 - 50)			
			0 - 50			
	00 0F	00aa aaaa	Category (0 - 50)			
			0 - 50			
	00 10	00aa aaaa	Category (0 - 50)			
			0 - 50			
#	00 11	0000 aaaa				
	00 12	0000 bbbb				
	00 13	0000 cccc				
	00 14	0000 dddd	EXZ (0 - 65535)			
#	00 15	0000 aaaa				
	00 16	0000 bbbb				
	00 17	0000 cccc				

Address		Description																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																			
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#	00 07	0000 bbbb	TVF Env T4 Velocity Sens	(28 - 228)	+-----+-----+-----+-----+			
	00 08	0000 aaaa		-100 - +100	* [Tone Common]			
	00 09	0000 bbbb			+-----+-----+-----+-----+			
	00 0A	0000 cccc			Offset			
#	00 0B	0000 dddd	TVF Env Time1	(0 - 1023)	Address			
	00 0C	0000 aaaa		0 - 1023	Description			
	00 0D	0000 bbbb			00 00	0aaa aaaa	Name	(32 - 127)
	00 0E	0000 cccc			00 01	0aaa aaaa	Name	32 - 127 [ASCII]
#	00 0F	0000 dddd	TVF Env Time2	(0 - 1023)	00 02	0aaa aaaa	Name	(32 - 127)
	00 10	0000 aaaa		0 - 1023	00 03	0aaa aaaa	Name	32 - 127 [ASCII]
	00 11	0000 bbbb			00 04	0aaa aaaa	Name	(32 - 127)
	00 12	0000 cccc			00 05	0aaa aaaa	Name	32 - 127 [ASCII]
#	00 13	0000 dddd	TVF Env Time3	(0 - 1023)	00 06	0aaa aaaa	Name	(32 - 127)
	00 14	0000 aaaa		0 - 1023	00 07	0aaa aaaa	Name	32 - 127 [ASCII]
	00 15	0000 bbbb			00 08	0aaa aaaa	Name	(32 - 127)
	00 16	0000 cccc			00 09	0aaa aaaa	Name	32 - 127 [ASCII]
#	00 17	0000 dddd	TVF Env Time4	(0 - 1023)	00 0A	0aaa aaaa	Name	(32 - 127)
	00 18	0000 aaaa		0 - 1023	00 0B	0aaa aaaa	Name	32 - 127 [ASCII]
	00 19	0000 bbbb			00 0C	0aaa aaaa	Name	(32 - 127)
	00 1A	0000 cccc			00 0D	0aaa aaaa	Name	32 - 127 [ASCII]
#	00 1B	0000 dddd	TVF Env Level0	(0 - 1023)	00 0E	0aaa aaaa	Name	(32 - 127)
	00 1C	0000 aaaa		0 - 1023	00 0F	0aaa aaaa	Name	32 - 127 [ASCII]
	00 1D	0000 bbbb			00 10	00aa aaaa	Category	(0 - 50)
	00 1E	0000 cccc			00 11	0000 aaaa		0 - 50
#	00 1F	0000 dddd	TVF Env Level1	(0 - 1023)	00 12	0000 bbbb		
	00 20	0000 aaaa		0 - 1023	00 13	0000 cccc		
	00 21	0000 bbbb			00 14	0000 dddd	Reserved	(0 - 0)
	00 22	0000 cccc			00 15	0aaa aaaa	Level	(0 - 127)
#	00 23	0000 dddd	TVF Env Level2	(0 - 1023)	00 16	0aaa aaaa	Pan	(0 - 127)
	00 24	0000 aaaa		0 - 1023	00 17	0000 000a	Priority	L64 - 63R
	00 25	0000 bbbb			00 18	0aaa aaaa	Tone Coarse Tune	(0 - 1)
	00 26	0000 cccc			00 19	0aaa aaaa	Tone Fine Tune	LAST, LOUDEST
#	00 27	0000 dddd	TVF Env Level3	(0 - 1023)	00 1A	0aaa aaaa	Octave Shift	(16 - 112)
	00 28	0000 aaaa		0 - 1023	00 1B	0000 00aa	Stretch Tune Depth	-48 - +48 [semitone]
	00 29	0000 bbbb			00 1C	0aaa aaaa	Analog Feel	(14 - 114)
	00 2A	0000 cccc			00 1D	0000 000a	Mono/Poly	-50 - +50 [cent]
#	00 2B	0000 dddd	TVF Env Level4	(0 - 1023)	00 1E	0000 000a	Legato Switch	(61 - 67)
	00 00 00 2C	Total Size		0 - 1023	00 1F	0000 aaaa	Legato Retrigger Interval	-3 - +3
					00 20	0000 000a	Portamento Switch	OFF, 1, 2, 3
					00 21	0000 000a	Portamento Mode	(0 - 127)
* [Partial Amp Env]					00 22	0000 000a	Portamento Type	0 - 127
+-----+-----+-----+-----+					00 23	0000 000a	Portamento Start	(0 - 1)
Offset					00 24	0000 aaaa		PITCH, NOTE
Address					00 25	0000 bbbb		
Description					00 26	0000 cccc		
#	00 00	0000 aaaa			00 27	0000 dddd	Portamento Time	(0 - 1023)
	00 01	0000 bbbb	TVA Env Time1 Velocity Sens	(28 - 228)	00 28	00aa aaaa	Bend Range Up	0 - 1023
	00 02	0000 aaaa		-100 - +100	00 29	00aa aaaa	Bend Range Down	(0 - 48)
	00 03	0000 bbbb	TVA Env Time4 Velocity Sens	(28 - 228)	00 2A	0000 000a	Bend Mode	0 - 48
#	00 04	0000 aaaa		-100 - +100	00 2B	0aaa aaaa	Soft Level Sens	(0 - 1)
	00 05	0000 bbbb			00 2C	0000 aaaa		NORMAL, CATCH+LAST
	00 06	0000 cccc			00 2D	0000 bbbb	Reserved	(0 - 100)
	00 07	0000 dddd	TVA Env Time1	(0 - 1023)	00 2E	0000 aaaa		0 - 100
#	00 08	0000 aaaa		0 - 1023	00 2F	0000 bbbb	Reserved	
	00 09	0000 bbbb			00 30	0000 aaaa		(0 - 0)
	00 0A	0000 cccc			00 31	0000 bbbb	Reserved	(0 - 0)
	00 0B	0000 dddd	TVA Env Time2	(0 - 1023)	00 32	0000 aaaa		(0 - 0)
#	00 0C	0000 aaaa		0 - 1023	00 33	0000 bbbb	Reserved	(0 - 0)
	00 0D	0000 bbbb			00 34	0000 00aa	Portamento Curve Type	(0 - 2)
	00 0E	0000 cccc			00 35	0000 000a	Partial LFO Phase Lock	1 - 3
	00 0F	0000 dddd	TVA Env Time3	(0 - 1023)				(0 - 1)
#	00 10	0000 aaaa		0 - 1023				
	00 11	0000 bbbb						
	00 12	0000 cccc						
	00 13	0000 dddd	TVA Env Time4	(0 - 1023)				
#	00 14	0000 aaaa		0 - 1023				
	00 15	0000 bbbb						
	00 16	0000 cccc						
	00 17	0000 dddd	TVA Env Level1	(0 - 1023)				
#	00 18	0000 aaaa		0 - 1023				
	00 19	0000 bbbb						
	00 1A	0000 cccc						
	00 1B	0000 dddd	TVA Env Level2	(0 - 1023)				
#	00 1C	0000 aaaa		0 - 1023				
	00 1D	0000 bbbb						
	00 1E	0000 cccc						
	00 1F	0000 dddd	TVA Env Level3	(0 - 1023)				
+-----+-----+-----+-----+								
00 00 00 20	Total Size							

JUNO-X MIDI Implementation

			OFF, ON
00 00 00 36	Total Size		
* [Tone PMT]			
Offset			
Address	Description		
00 00	0000 00aa	Velocity Control	(0 - 3)
00 01	0000 000a	PMT Level Curve	OFF, ON, RANDOM, CYCLE (0 - 1)
00 02	0000 000a	Partial Switch	EXP, LINEAR (0 - 1)
00 03	0aaa aaaa	KeyRange Lower	OFF, ON (0 - 127)
00 04	0aaa aaaa	KeyRange Upper	0 - 127 (0 - 127)
00 05	0aaa aaaa	KeyFade Lower	0 - 127 (0 - 127)
00 06	0aaa aaaa	KeyFade Upper	0 - 127 (0 - 127)
00 07	0aaa aaaa	VeloRange Lower	0 - 127 (1 - 127)
00 08	0aaa aaaa	VeloRange Upper	1 - 127 (1 - 127)
00 09	0aaa aaaa	VeloFade Lower	1 - 127 (0 - 127)
00 0A	0aaa aaaa	VeloFade Upper	0 - 127 (0 - 127)
00 0B	0000 000a	Partial Switch	0 - 127 (0 - 1)
00 0C	0aaa aaaa	KeyRange Lower	OFF, ON (0 - 127)
00 0D	0aaa aaaa	KeyRange Upper	0 - 127 (0 - 127)
00 0E	0aaa aaaa	KeyFade Lower	0 - 127 (0 - 127)
00 0F	0aaa aaaa	KeyFade Upper	0 - 127 (0 - 127)
00 10	0aaa aaaa	VeloRange Lower	0 - 127 (1 - 127)
00 11	0aaa aaaa	VeloRange Upper	1 - 127 (1 - 127)
00 12	0aaa aaaa	VeloFade Lower	1 - 127 (0 - 127)
00 13	0aaa aaaa	VeloFade Upper	0 - 127 (0 - 127)
00 14	0000 000a	Partial Switch	0 - 127 (0 - 1)
00 15	0aaa aaaa	KeyRange Lower	OFF, ON (0 - 127)
00 16	0aaa aaaa	KeyRange Upper	0 - 127 (0 - 127)
00 17	0aaa aaaa	KeyFade Lower	0 - 127 (0 - 127)
00 18	0aaa aaaa	KeyFade Upper	0 - 127 (0 - 127)
00 19	0aaa aaaa	VeloRange Lower	0 - 127 (1 - 127)
00 1A	0aaa aaaa	VeloRange Upper	1 - 127 (1 - 127)
00 1B	0aaa aaaa	VeloFade Lower	1 - 127 (0 - 127)
00 1C	0aaa aaaa	VeloFade Upper	0 - 127 (0 - 127)
00 1D	0000 000a	Partial Switch	0 - 127 (0 - 1)
00 1E	0aaa aaaa	KeyRange Lower	OFF, ON (0 - 127)
00 1F	0aaa aaaa	KeyRange Upper	0 - 127 (0 - 127)
00 20	0aaa aaaa	KeyFade Lower	0 - 127 (0 - 127)
00 21	0aaa aaaa	KeyFade Upper	0 - 127 (0 - 127)
00 22	0aaa aaaa	VeloRange Lower	0 - 127 (1 - 127)
00 23	0aaa aaaa	VeloRange Upper	1 - 127 (1 - 127)
00 24	0aaa aaaa	VeloFade Lower	1 - 127 (0 - 127)
00 25	0aaa aaaa	VeloFade Upper	0 - 127 (0 - 127)
00 00 00 26	Total Size		

* [Tone Partial]

Offset	Address	Description	
00 00	0aaa aaaa	Level	(0 - 127)
00 01	0aaa aaaa	Coarse Tune	(16 - 112)
			-48 - 48

00 02	0aaa aaaa	Fine Tune	(14 - 114)
			-50 - 50 [cent]
00 03	0000 aaaa		
00 04	0000 bbbb		
00 05	0000 cccc		
00 06	0000 dddd	Random Pitch Depth	(0 - 1200)
			0 - 1200 [cent]
00 07	0aaa aaaa	Pan	(0 - 127)
			L64 - 63R
00 08	0000 aaaa		
00 09	0000 bbbb	Pan Keyfollow	(28 - 228)
			-100 - +100
00 0A	00aa aaaa	Random Pan Depth	(0 - 63)
			0 - 63
00 0B	0aaa aaaa	Alternate Pan Depth	(0 - 127)
			L64 - 63R
00 0C	0000 000a	Envelope Mode	(0 - 1)
			NO-SUS, SUSTAIN
00 0D	0000 00aa	Delay Mode	(0 - 3)
			NORMAL, HOLD, KEYOFF-NORMAL, KEYOFF-DECAY
00 0E	0000 000a	DelayTime Sync	(0 - 1)
			OFF, ON
00 0F	000a aaaa	DelayTime (note)	(0 - 21)
			1/64T, 1/64,1/32T,1/32,1/16T,1/32.,1/16,1/8T,1/16.,1/8, 1/4T,1/8.,1/4,1/2T,1/4.,1/2,1T,1/2.,1,2T,1.,2
00 10	0000 aaaa		
00 11	0000 bbbb		
00 12	0000 cccc		
00 13	0000 dddd	DelayTime	(0 - 1023)
			0 - 1023
00 14	0aaa aaaa	Chorus Send Level	(0 - 127)
			0 - 127
00 15	0aaa aaaa	Reverb Send Level	(0 - 127)
			0 - 127
00 16	0000 000a	Receive Bender	(0 - 1)
			OFF, ON
00 17	0000 000a	Receive Expression	(0 - 1)
			OFF, ON
00 18	0000 000a	Receive Hold-1	(0 - 1)
			OFF, ON
00 19	0000 000a	Redamper Switch	(0 - 1)
			OFF, ON
00 1A	0000 000a	Output Assign	(0 - 1)
			DRY, MFX
00 1B	0000 00aa	Wave Group Type	(0 - 3)
			INT, EXP, SAMP, MSAMP
00 1C	0000 aaaa		
00 1D	0000 bbbb		
00 1E	0000 cccc		
00 1F	0000 dddd	Wave Group ID	(0 - 16383)
			0 - 16383
00 20	0000 aaaa		
00 21	0000 bbbb		
00 22	0000 cccc		
00 23	0000 dddd	Wave Number L	(0 - 16383)
			0 - 16383
00 24	0000 aaaa		
00 25	0000 bbbb		
00 26	0000 cccc		
00 27	0000 dddd	Wave Number R	(0 - 16383)
			0 - 16383
00 28	0000 0aaa	Gain	(0 - 5)
			-18, -12, -6, 0, +6, +12[dB]
00 29	0000 000a	FXM Switch	(0 - 1)
			OFF, ON
00 2A	0000 0aaa	FXM Color	(1 - 4)
			1 - 4
00 2B	000a aaaa	FXM Depth	(0 - 16)
			0 - 16
00 2C	0000 aaaa		
00 2D	0000 bbbb		
00 2E	0000 cccc		
00 2F	0000 dddd	Pitch Keyfollow	(824 - 1224)
			-200 - +200
00 30	0aaa aaaa	Soft EQ Sens	(0 - 100)
			0 - 100
00 31	0000 0aaa	TVF Filter Type	(0 - 6)
			OFF, LPF, BPF, HPF, PKG, LPF2, LPF3
00 32	0000 aaaa		
00 33	0000 bbbb		
00 34	0000 cccc		
00 35	0000 dddd	TVF Cutoff Frequency	(0 - 1023)
			0 - 1023
00 36	0000 aaaa		
00 37	0000 bbbb		
00 38	0000 cccc		
00 39	0000 dddd	TVF Cutoff Keyfollow	(824 - 1224)
			-200 - +200
00 3A	0000 0aaa	TVF Cutoff Velocity Curve	(0 - 7)
			0 - 7
00 3B	0000 aaaa		
00 3C	0000 bbbb	TVF Cutoff Velocity Sens	(28 - 228)
			-100 - +100
00 3D	0000 aaaa		
00 3E	0000 bbbb		
00 3F	0000 cccc		
00 40	0000 dddd	TVF Resonance	(0 - 1023)
			0 - 1023

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			MFX-CTRL3, MFX-CTRL4, PW, PWM, FAT, XMOD, LFO1_STEP, LFO2_STEP, SSAW-DETN, PIT_DEPTH, TVF-DEPTH, TVA-DEPTH, XMOD2, ATT, RING-OSC1-LEV, RING-OSC2-LEV, XMOD-OSC1-LEV, XMOD-OSC2-LEV	00 00 01 00 Total Size	
00 6E	0aaa aaaa	Sens 3	(1 - 127) -63 - +63	* [Tone Synth Common]	
00 6F	00aa aaaa	Destination 4	(0 - 48)	Offset	Description
		OFF, PCH, CUT, RES, LEV, PAN, CHO, REV, PIT-LFO1, PIT-LFO2, TVF-LFO1, TVF-LFO2, TVA-LFO1, TVA-LFO2, PAN-LFO1, PAN-LFO2, LFO1-RATE, LFO2-RATE, PIT-ATK, PIT-DCY, PIT-REL, TVF-ATK, TVF-DCY, TVF-REL, TVA-ATK, TVA-DCY, TVA-REL, PMT, FXM, MFX-CTRL1, MFX-CTRL2, MFX-CTRL3, MFX-CTRL4, PW, PWM, FAT, XMOD, LFO1_STEP, LFO2_STEP, SSAW-DETN, PIT_DEPTH, TVF-DEPTH, TVA-DEPTH, XMOD2, ATT, RING-OSC1-LEV, RING-OSC2-LEV, XMOD-OSC1-LEV, XMOD-OSC2-LEV		00 00	0000 000a Unison Switch (0 - 1)
				00 01	0000 aaaa Unison Size (2 - 8)
				00 02	0aaa aaaa Unison Detune (0 - 100)
				00 03	0aaa aaaa Bend Range Fine Up (0 - 100)
				00 04	0aaa aaaa Bend Range Fine Down (0 - 100)
00 70	0aaa aaaa	Sens 4	(1 - 127) -63 - +63	# 00 05	0000 aaaa Pitch Drift (0 - 255)
00 71	0aaa aaaa	Source	(0 - 108)	00 06	0000 bbbb Pitch Drift Cycle Number (0 - 8)
		OFF, CC01, CC02, CC03, CC04, CC05, CC06, CC07, CC08, CC09, CC10, CC11, CC12, CC13, CC14, CC15, CC16, CC17, CC18, CC19, CC20, CC21, CC22, CC23, CC24, CC25, CC26, CC27, CC28, CC29, CC30, CC31, CC33, CC34, CC35, CC36, CC37, CC38, CC39, CC40, CC41, CC42, CC43, CC44, CC45, CC46, CC47, CC48, CC49, CC50, CC51, CC52, CC53, CC54, CC55, CC56, CC57, CC58, CC59, CC60, CC61, CC62, CC63, CC64, CC65, CC66, CC67, CC68, CC69, CC70, CC71, CC72, CC73, CC74, CC75, CC76, CC77, CC78, CC79, CC80, CC81, CC82, CC83, CC84, CC85, CC86, CC87, CC88, CC89, CC90, CC91, CC92, CC93, CC94, CC95, BEND, APT, SYS-CTRL1, SYS-CTRL2, SYS-CTRL3, SYS-CTRL4, VELOCITY, KEYFOLLOW, TEMPO, LFO1, LFO2, PIT-ENV, TVF-ENV, TVA-ENV		00 07	0000 aaaa Condition (0 - 100)
				00 08	0aaa aaaa Condition (0 - 100)
				00 00 00 09 Total Size	
				* [Tone Synth PMT]	
00 72	00aa aaaa	Destination 1	(0 - 48)	Offset	Description
		OFF, PCH, CUT, RES, LEV, PAN, CHO, REV, PIT-LFO1, PIT-LFO2, TVF-LFO1, TVF-LFO2, TVA-LFO1, TVA-LFO2, PAN-LFO1, PAN-LFO2, LFO1-RATE, LFO2-RATE, PIT-ATK, PIT-DCY, PIT-REL, TVF-ATK, TVF-DCY, TVF-REL, TVA-ATK, TVA-DCY, TVA-REL, PMT, FXM, MFX-CTRL1, MFX-CTRL2, MFX-CTRL3, MFX-CTRL4, PW, PWM, FAT, XMOD, LFO1_STEP, LFO2_STEP, SSAW-DETN, PIT_DEPTH, TVF-DEPTH, TVA-DEPTH, XMOD2, ATT, RING-OSC1-LEV, RING-OSC2-LEV, XMOD-OSC1-LEV, XMOD-OSC2-LEV		00 00	0000 0aaa Structure1-2 (0 - 4)
				00 01	0000 0aaa Structure3-4 (0 - 4)
				00 02	0aaa aaaa RING1-2 Level (0 - 127)
				00 03	0aaa aaaa RING3-4 Level (0 - 127)
				00 04	0aaa aaaa RING OSC1 Level (0 - 127)
00 73	0aaa aaaa	Sens 1	(1 - 127) -63 - +63	00 05	0aaa aaaa RING OSC2 Level (0 - 127)
00 74	00aa aaaa	Destination 2	(0 - 48)	00 06	0aaa aaaa RING OSC3 Level (0 - 127)
		OFF, PCH, CUT, RES, LEV, PAN, CHO, REV, PIT-LFO1, PIT-LFO2, TVF-LFO1, TVF-LFO2, TVA-LFO1, TVA-LFO2, PAN-LFO1, PAN-LFO2, LFO1-RATE, LFO2-RATE, PIT-ATK, PIT-DCY, PIT-REL, TVF-ATK, TVF-DCY, TVF-REL, TVA-ATK, TVA-DCY, TVA-REL, PMT, FXM, MFX-CTRL1, MFX-CTRL2, MFX-CTRL3, MFX-CTRL4, PW, PWM, FAT, XMOD, LFO1_STEP, LFO2_STEP, SSAW-DETN, PIT_DEPTH, TVF-DEPTH, TVA-DEPTH, XMOD2, ATT, RING-OSC1-LEV, RING-OSC2-LEV, XMOD-OSC1-LEV, XMOD-OSC2-LEV		00 07	0aaa aaaa RING OSC4 Level (0 - 127)
				# 00 08	0000 aaaa
				00 09	0000 bbbb
				00 0A	0000 cccc
00 75	0aaa aaaa	Sens 2	(1 - 127) -63 - +63	00 0B	0000 dddd CrossMod1-2 Depth (0 - 10800)
00 76	00aa aaaa	Destination 3	(0 - 48)	# 00 0C	0000 aaaa
		OFF, PCH, CUT, RES, LEV, PAN, CHO, REV, PIT-LFO1, PIT-LFO2, TVF-LFO1, TVF-LFO2, TVA-LFO1, TVA-LFO2, PAN-LFO1, PAN-LFO2, LFO1-RATE, LFO2-RATE, PIT-ATK, PIT-DCY, PIT-REL, TVF-ATK, TVF-DCY, TVF-REL, TVA-ATK, TVA-DCY, TVA-REL, PMT, FXM, MFX-CTRL1, MFX-CTRL2, MFX-CTRL3, MFX-CTRL4, PW, PWM, FAT, XMOD, LFO1_STEP, LFO2_STEP, SSAW-DETN, PIT_DEPTH, TVF-DEPTH, TVA-DEPTH, XMOD2, ATT, RING-OSC1-LEV, RING-OSC2-LEV, XMOD-OSC1-LEV, XMOD-OSC2-LEV		00 0D	0000 bbbb
				00 0E	0000 cccc
				00 0F	0000 dddd CrossMod3-4 Depth (0 - 10800)
				00 10	0aaa aaaa CrossMod OSC1 Level (0 - 127)
				00 11	0aaa aaaa CrossMod OSC2 Level (0 - 127)
				00 12	0aaa aaaa CrossMod OSC3 Level (0 - 127)
00 77	0aaa aaaa	Sens 3	(1 - 127) -63 - +63	00 13	0aaa aaaa CrossMod OSC4 Level (0 - 127)
00 78	00aa aaaa	Destination 4	(0 - 48)	00 14	0000 000a Partial Phase Lock (0 - 1)
		OFF, PCH, CUT, RES, LEV, PAN, CHO, REV, PIT-LFO1, PIT-LFO2, TVF-LFO1, TVF-LFO2, TVA-LFO1, TVA-LFO2, PAN-LFO1, PAN-LFO2, LFO1-RATE, LFO2-RATE, PIT-ATK, PIT-DCY, PIT-REL, TVF-ATK, TVF-DCY, TVF-REL, TVA-ATK, TVA-DCY, TVA-REL, PMT, FXM, MFX-CTRL1, MFX-CTRL2, MFX-CTRL3, MFX-CTRL4, PW, PWM, FAT, XMOD, LFO1_STEP, LFO2_STEP, SSAW-DETN, PIT_DEPTH, TVF-DEPTH, TVA-DEPTH, XMOD2, ATT, RING-OSC1-LEV, RING-OSC2-LEV, XMOD-OSC1-LEV, XMOD-OSC2-LEV		00 15	0aaa aaaa CrossMod2 1-2 Depth (0 - 127)
				00 16	0aaa aaaa CrossMod2 3-4 Depth (0 - 127)
				00 00 00 17 Total Size	
00 79	0aaa aaaa	Sens 4	(1 - 127) -63 - +63	* [Tone Synth Partial]	
00 7A	0aaa aaaa	Damper Free Note	(0 - 127)	Offset	Description
		OFF, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127		00 00	0000 0aaa OSC Type (0 - 4)
				00 01	0000 aaaa VA Waveform (0 - 8)
				# 00 02	0000 aaaa SAW, SQR, TRI, SIN, RAMP, JUNO, TRI2, TRI3, SIN2
				00 03	0000 bbbb
				00 04	0000 cccc
00 7B	0aaa aaaa	Stereo Width	(0 - 100)	00 05	0000 dddd PCM-Sync Wave Number (0 - 16383)
00 7C	0aaa aaaa	Stereo Detune	(14 - 114) -50 - +50 [cent]	00 06	0aaa aaaa Pulse Width (0 - 127)
# 00 7D	0000 aaaa			00 07	0aaa aaaa PWM Depth (1 - 127)
00 7E	0000 bbbb	Damper Free Decay Offset	(28 - 228) -100 - +100	00 08	0aaa aaaa SuperSAW Detune (0 - 127)
00 7F	0000 000a	Wave Tempo Sync	(0 - 1) OFF, ON	00 09	0000 00aa Click Type (0 - 3)

#	00 0A	0000 aaaa	SOFT, HARD, NATURAL, OFF	00 24	0000 dddd	OSC2 WAVE FORM	(0 - 27)
	00 0B	0000 bbbb				PW, SAW, SUB1OCT1, SUB2OCT1, SUB2OCT2, NOISE,	
	00 0C	0000 cccc				PW, SAW, SUB1OCT1, NOISE, SIN, TRI, SAW, PW,	
	00 0D	0000 dddd	HPF Cutoff			SQR, NOISE, SAW, PULSE, SQR, NOISE, TRI, SAW+TRI,	
			(0 - 1023)			SAW, SQR, SQR2, SQR3, WHITE NOISE, PINK NOISE	
	00 0E	0000 000a	Filter Type	00 25	0000 0aaa	OSC2 FEET	(0 - 4)
			(0 - 1)			32', 16', 8', 4', 2'	
	00 0F	0000 00aa	Filter Slope	00 26	0aaa aaaa	OSC2 COARSE TUNE	(16 - 112)
			TVF, VCF			-48 - 48	
			(0 - 2)	00 27	0aaa aaaa	OSC2 FINE TUNE	(14 - 114)
	00 10	0000 000a	ADSR Env Switch			-50 - 50	
			(0 - 1)	00 28	0aaa aaaa	OSC2 LOW FREQ PITCH	(0 - 127)
	00 11	0aaa aaaa	Fat	00 29	0000 000a	OSC2 MODE	(0 - 1)
			(0 - 127)			NORMAL, LO FREQ	
	00 12	0000 00aa	VCF Type	00 2A	0aaa aaaa	OSC2 PULSE WIDTH	(0 - 127)
			(0 - 3)	00 2B	0000 aaaa		
			1,2,3,4	00 2C	0000 bbbb	OSC3 MOD	(28 - 228)
#	00 13	0000 aaaa				-100 - 100	
	00 14	0000 bbbb	OSC Attenuator	#	00 2D	0000 aaaa	
			(0 - 255)		00 2E	0000 bbbb	OSC3 PITCH ENV DEPTH
	00 15	0aaa aaaa	Cutoff Keyfollow Base Point			-100 - 100	
			(0 - 127)	#	00 2F	0000 aaaa	
	00 16	0000 000a	VA Waveform Invert Switch		00 30	0000 bbbb	
			(0 - 1)		00 31	0000 cccc	
	00 17	0aaa aaaa	TVF Env Fine Depth		00 32	0000 dddd	OSC3 WAVE FORM
			(1 - 127)				(0 - 27)
	00 18	0000 000a	PENV LFO Trigger Switch				PW, SAW, SQR 1OCT, SQR 2OCT, SH SubOSC, NOISE,
			(0 - 1)				PW, SAW, SQR 1OCT, NOISE, SIN, TRI, SAW, PW,
	00 19	0000 000a	FENV LFO Trigger Switch				SQR, NOISE, SAW, PULSE, SQR, NOISE, TRI, SAW+TRI,
			(0 - 1)				SAW, SQR, SQR2, SQR3, WHITE NOISE, PINK NOISE
	00 1A	0000 000a	AENV LFO Trigger Switch	00 33	0000 0aaa	OSC3 FEET	(0 - 4)
			(0 - 1)			32', 16', 8', 4', 2'	
	00 1B	0aaa aaaa	Pitch Down Depth	00 34	0aaa aaaa	OSC3 COARSE TUNE	(16 - 112)
			(0 - 120)			-48 - 48	
	00 1C	0aaa aaaa	VCF Gain Correction	00 35	0aaa aaaa	OSC3 FINE TUNE	(14 - 114)
			(0 - 127)			-50 - 50	
			0 - 127	00 36	0aaa aaaa	OSC3 LOW FREQ PITCH	(0 - 127)
	00 00 00 1D	Total Size		00 37	0000 000a	OSC3 MODE	(0 - 1)
						NORMAL, LO FREQ	
				00 38	0000 000a	NOISE	(0 - 1)
						WHITE, PINK	
				#	00 39	0000 aaaa	
					00 3A	0000 bbbb	
					00 3B	0000 cccc	
					00 3C	0000 dddd	MOD DEPTH
					00 3D	0000 00aa	MOD MODE
							(0 - 10800)
							(0 - 3)
							OFF, SYNC, RING, X-MOD
				#	00 3E	0000 aaaa	
					00 3F	0000 bbbb	OSC1 LEVEL
					00 40	0000 aaaa	(0 - 255)
				#	00 41	0000 bbbb	OSC2 LEVEL
					00 42	0000 aaaa	(0 - 255)
				#	00 43	0000 bbbb	OSC3 LEVEL
					00 44	0000 aaaa	(0 - 255)
				#	00 45	0000 bbbb	NOISE LEVEL
					00 46	0aaa aaaa	MIXER ENV DEPTH
							(1 - 127)
							-63 - 63
					00 47	0000 000a	MIXER ENV SELECT
							(0 - 1)
							1, 2
					00 48	0000 00aa	MIXER DYNAMICS
					00 49	0000 00aa	FILTER TYPE
					00 4A	0000 00aa	FILTER HPF STEP
							(0 - 3)
				#	00 4B	0000 aaaa	
					00 4C	0000 bbbb	
					00 4D	0000 cccc	
					00 4E	0000 dddd	FILTER HPF
							(0 - 1023)
				#	00 4F	0000 aaaa	
					00 50	0000 bbbb	
					00 51	0000 cccc	
					00 52	0000 dddd	FILTER FREQ
							(0 - 1023)
				#	00 53	0000 aaaa	
					00 54	0000 bbbb	
					00 55	0000 cccc	
					00 56	0000 dddd	FILTER RESO
							(0 - 1023)
				#	00 57	0000 aaaa	
					00 58	0000 bbbb	
					00 59	0000 cccc	
					00 5A	0000 dddd	FILTER ENV DEPTH
							(1 - 2047)
							-1023 - 1023
					00 5B	0000 000a	FILTER ENV SELECT
							(0 - 1)
							1, 2
					00 5C	0000 00aa	FILTER SLOPE
							(0 - 2)
							-12dB, -18dB, -24dB
				#	00 5D	0000 aaaa	
					00 5E	0000 bbbb	
					00 5F	0000 cccc	
					00 60	0000 dddd	FILTER KEY FOLLOW
							(824 - 1224)
							-200 - 200
				#	00 61	0000 aaaa	
					00 62	0000 bbbb	FILTER MOD
							(28 - 228)
							-100 - 100
					00 63	0aaa aaaa	AMP LEVEL
							(0 - 127)
				#	00 64	0000 aaaa	
					00 65	0000 bbbb	AMP MOD
							(-100 - 100)
					00 66	0000 00aa	AMP MOD STEP
							(0 - 3)
					00 67	0000 00aa	AMP ENV SELECT
							(0 - 3)
							ENV F&A, ENV2, G-AMP, G-ENV2
				#	00 68	0000 aaaa	
					00 69	0000 bbbb	
					00 6A	0000 cccc	

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#	00 6B	0000 dddd	ENV1 ATTACK	(0 - 1023)	#	00 0B	0aaa aaaa	LFO PITCH DEPTH	(0 - 100)
	00 6C	0000 aaaa				00 0C	0aaa aaaa	LFO FILTER DEPTH	(0 - 100)
	00 6D	0000 bbbb				00 0D	0aaa aaaa	LFO AMP DEPTH	(0 - 100)
	00 6E	0000 cccc				00 0E	0000 000a	SUPER SAW SWITCH	(0 - 1)
	00 6F	0000 dddd	ENV1 DECAY	(0 - 1023)				OFF, ON	
#	00 70	0000 aaaa				00 0F	0000 00aa	PW MODE	(0 - 2)
	00 71	0000 bbbb						LFO, MANUAL, ENV	
	00 72	0000 cccc				00 10	0aaa aaaa	PW PWM (SSAW DETUNE)	(0 - 127)
	00 73	0000 dddd	ENV1 SUSTAIN	(0 - 1023)		00 11	0000 000a	SUB OSC PHASE SHIFT	(0 - 1)
#	00 74	0000 aaaa						OFF, ON	
	00 75	0000 bbbb				00 12	0aaa aaaa	PITCH	(52 - 76)
	00 76	0000 cccc						-12 - +12 [semitone]	
	00 77	0000 dddd	ENV1 RELEASE	(0 - 1023)		00 13	0aaa aaaa	OSC DETUNE	(14 - 114)
#	00 78	0000 aaaa						-50 - 50	
	00 79	0000 bbbb	ENV1 KEY FOLLOW	(28 - 228)		00 14	0aaa aaaa	OSC PAN SPLIT	(14 - 114)
				-100 - 100				-50 - 50	
	00 7A	0000 00aa	ENV1 KEY FOLLOW STEP	(0 - 3)	#	00 15	0000 aaaa		
#	00 7B	0000 aaaa				00 16	0000 bbbb	PW LEVEL (SSAW LEVEL)	(0 - 255)
	00 7C	0000 bbbb			#	00 17	0000 aaaa		
	00 7D	0000 cccc				00 18	0000 bbbb	SAW LEVEL	(0 - 255)
	00 7E	0000 dddd	ENV2 ATTACK	(0 - 1023)	#	00 19	0000 aaaa		
#	00 7F	0000 aaaa				00 1A	0000 bbbb	SUB OSC LEVEL	(0 - 255)
	01 00	0000 bbbb			#	00 1B	0000 aaaa		
	01 01	0000 cccc				00 1C	0000 bbbb	NOISE LEVEL	(0 - 255)
	01 02	0000 dddd	ENV2 DECAY	(0 - 1023)		00 1D	0aaa aaaa	AMP LEVEL	(0 - 127)
#	01 03	0000 aaaa				00 1E	0000 000a	AMP CONTROL SELECT	(0 - 1)
	01 04	0000 bbbb						ENV, GATE	
	01 05	0000 cccc			#	00 1F	0000 aaaa		
	01 06	0000 dddd	ENV2 SUSTAIN	(0 - 1023)		00 20	0000 bbbb	AMP VELOCITY SENS	(28 - 228)
#	01 07	0000 aaaa						-100 - 100	
	01 08	0000 bbbb			#	00 21	0000 aaaa		
	01 09	0000 cccc				00 22	0000 bbbb		
	01 0A	0000 dddd	ENV2 RELEASE	(0 - 1023)		00 23	0000 cccc		
	01 0B	0000 00aa	ENV2 MODE	(0 - 2)		00 24	0000 dddd	FILTER CUTOFF FREQ	(0 - 1023)
				GATE+TRIG, GATE, LFO	#	00 25	0000 aaaa		
#	01 0C	0000 aaaa				00 26	0000 bbbb		
	01 0D	0000 bbbb	ENV2 KEY FOLLOW	(28 - 228)		00 27	0000 cccc		
				-100 - 100		00 28	0000 dddd	FILTER RESONANCE	(0 - 1023)
	01 0E	0000 00aa	ENV2 KEY FOLLOW STEP	(0 - 3)	#	00 29	0000 aaaa		
	01 0F	0000 00aa	DYNAMICS PITCH	(0 - 3)		00 2A	0000 bbbb		
	01 10	0000 00aa	DYNAMICS FILTER	(0 - 3)		00 2B	0000 cccc		
	01 11	0000 00aa	DYNAMICS AMP	(0 - 3)		00 2C	0000 dddd	FILTER ENV DEPTH	(1 - 2047)
#	01 12	0000 aaaa						-1023 - 1023	
	01 13	0000 bbbb	PITCH DRIFT	(0 - 255)	#	00 2D	0000 aaaa		
	01 14	0000 00aa	BENDER PITCH DEPTH STEP	(0 - 3)		00 2E	0000 bbbb		
				2, 3, 4, 7		00 2F	0000 cccc		
	01 15	0aaa aaaa	BENDER FILTER DEPTH	(1 - 127)		00 30	0000 dddd	FILTER KEY FOLLOW	(824 - 1224)
				-63 - 63				-200 - 200	
	01 16	0aaa aaaa	MODULATION LFO DEPTH	(1 - 127)		00 31	0000 00aa	FILTER TYPE	(0 - 2)
				-63 - 63				R, M, S	
#	01 17	0000 aaaa			#	00 32	0000 aaaa		
	01 18	0000 bbbb				00 33	0000 bbbb	FILTER VELOCITY SENS	(28 - 228)
	01 19	0000 cccc						-100 - 100	
	01 1A	0000 dddd	BENDER PITCH DEPTH	(0 - 4800)	#	00 34	0000 aaaa		
#	01 1B	0000 aaaa				00 35	0000 bbbb		
	01 1C	0000 bbbb				00 36	0000 cccc		
	01 1D	0000 cccc				00 37	0000 dddd	PITCH ENV ATTACK	(0 - 1023)
	01 1E	0000 dddd	PORTAMENT TIME	(0 - 1023)	#	00 38	0000 aaaa		
	01 1F	0000 00aa	PORTAMENT MODE	(0 - 2)		00 39	0000 bbbb		
				OFF, ON, AUTO		00 3A	0000 cccc		
	01 20	0000 00aa	KEY MODE	(0 - 3)		00 3B	0000 dddd	PITCH ENV DECAY	(0 - 1023)
				POLY, SOLO, UNISON, SI-UNISON	#	00 3C	0000 aaaa		
	01 21	0000 000a	PARAMETER EXPANSION SWITCH	(0 - 1)		00 3D	0000 bbbb		
				OFF, ON		00 3E	0000 cccc		
	01 22	0aaa aaaa	CONDITION	(0 - 100)		00 3F	0000 dddd	PITCH ENV SUSTAIN	(0 - 1023)
				0 - 100	#	00 40	0000 aaaa		
	01 23	0000 00aa	PORTAMENT CURVE TYPE	(0 - 3)		00 41	0000 bbbb		
				ORIGINAL, LINER, EXP1, EXP2		00 42	0000 cccc		
	01 24	0aaa aaaa	AFTER TOUCH SENS LFO	(1 - 127)		00 43	0000 dddd	PITCH ENV RELEASE	(0 - 1023)
				-63 - 63	#	00 44	0000 aaaa		
	01 25	0aaa aaaa	AFTER TOUCH SENS FREQ	(1 - 127)		00 45	0000 bbbb	PITCH ENV VELOCITY SENS	(28 - 228)
				-63 - 63				-100 - 100	
	01 26	0aaa aaaa	AFTER TOUCH SENS LEVEL	(1 - 127)	#	00 46	0000 aaaa		
				-63 - 63		00 47	0000 bbbb	PITCH ENV DEPTH	(28 - 228)
								-100 - 100	
00 00 01 27 Total Size					#	00 48	0000 aaaa		
						00 49	0000 bbbb		
						00 4A	0000 cccc		
						00 4B	0000 dddd	ENV ATTACK	(0 - 1023)
					#	00 4C	0000 aaaa		
						00 4D	0000 bbbb		
						00 4E	0000 cccc		
						00 4F	0000 dddd	ENV DECAY	(0 - 1023)
					#	00 50	0000 aaaa		
						00 51	0000 bbbb		
						00 52	0000 cccc		
					#	00 53	0000 dddd	ENV SUSTAIN	(0 - 1023)
						00 54	0000 aaaa		
						00 55	0000 bbbb		
						00 56	0000 cccc		
						00 57	0000 dddd	ENV RELEASE	(0 - 1023)
						00 58	0aaa aaaa	AFTER TOUCH SENS LFO	(1 - 127)
					#			-63 - 63	
						00 59	0aaa aaaa	AFTER TOUCH SENS FREQ	(1 - 127)
								-63 - 63	
						00 5A	0aaa aaaa	AFTER TOUCH SENS LEVEL	(1 - 127)
								-63 - 63	
						00 5B	0aaa aaaa	BENDER FILTER DEPTH	(1 - 127)
								-63 - 63	

[illegible]

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Offset	Address	Description		#	00 4C	0000 aaaa		
					00 4D	0000 bbbb		
					00 4E	0000 cccc		
	00 00	0000 000a	Switch (0 - 1)		00 4F	0000 dddd	Chorus Parameter 19	(12768 - 52768)
			OFF, ON	#	00 50	0000 aaaa		-20000 - 20000
	00 01	0000 aaaa	Chorus Type (0 - 10)		00 51	0000 bbbb		
	00 02	0aaa aaaa	Chorus Level (0 - 127)		00 52	0000 cccc		
	00 03	0aaa aaaa	Reverb Send Level (0 - 127)		00 53	0000 dddd	Chorus Parameter 20	(12768 - 52768)
#	00 04	0000 aaaa						-20000 - 20000
	00 05	0000 bbbb						
	00 06	0000 cccc			00 00 00 54	Total Size		
	00 07	0000 dddd	Chorus Parameter 1 (12768 - 52768)					
			-20000 - 20000					
#	00 08	0000 aaaa						
	00 09	0000 bbbb						
	00 0A	0000 cccc						
	00 0B	0000 dddd	Chorus Parameter 2 (12768 - 52768)					
			-20000 - 20000					
#	00 0C	0000 aaaa						
	00 0D	0000 bbbb						
	00 0E	0000 cccc						
	00 0F	0000 dddd	Chorus Parameter 3 (12768 - 52768)					
			-20000 - 20000					
#	00 10	0000 aaaa						
	00 11	0000 bbbb						
	00 12	0000 cccc						
	00 13	0000 dddd	Chorus Parameter 4 (12768 - 52768)					
			-20000 - 20000					
#	00 14	0000 aaaa						
	00 15	0000 bbbb						
	00 16	0000 cccc						
	00 17	0000 dddd	Chorus Parameter 5 (12768 - 52768)					
			-20000 - 20000					
#	00 18	0000 aaaa						
	00 19	0000 bbbb						
	00 1A	0000 cccc						
	00 1B	0000 dddd	Chorus Parameter 6 (12768 - 52768)					
			-20000 - 20000					
#	00 1C	0000 aaaa						
	00 1D	0000 bbbb						
	00 1E	0000 cccc						
	00 1F	0000 dddd	Chorus Parameter 7 (12768 - 52768)					
			-20000 - 20000					
#	00 20	0000 aaaa						
	00 21	0000 bbbb						
	00 22	0000 cccc						
	00 23	0000 dddd	Chorus Parameter 8 (12768 - 52768)					
			-20000 - 20000					
#	00 24	0000 aaaa						
	00 25	0000 bbbb						
	00 26	0000 cccc						
	00 27	0000 dddd	Chorus Parameter 9 (12768 - 52768)					
			-20000 - 20000					
#	00 28	0000 aaaa						
	00 29	0000 bbbb						
	00 2A	0000 cccc						
	00 2B	0000 dddd	Chorus Parameter 10 (12768 - 52768)					
			-20000 - 20000					
#	00 2C	0000 aaaa						
	00 2D	0000 bbbb						
	00 2E	0000 cccc						
	00 2F	0000 dddd	Chorus Parameter 11 (12768 - 52768)					
			-20000 - 20000					
#	00 30	0000 aaaa						
	00 31	0000 bbbb						
	00 32	0000 cccc						
	00 33	0000 dddd	Chorus Parameter 12 (12768 - 52768)					
			-20000 - 20000					
#	00 34	0000 aaaa						
	00 35	0000 bbbb						
	00 36	0000 cccc						
	00 37	0000 dddd	Chorus Parameter 13 (12768 - 52768)					
			-20000 - 20000					
#	00 38	0000 aaaa						
	00 39	0000 bbbb						
	00 3A	0000 cccc						
	00 3B	0000 dddd	Chorus Parameter 14 (12768 - 52768)					
			-20000 - 20000					
#	00 3C	0000 aaaa						
	00 3D	0000 bbbb						
	00 3E	0000 cccc						
	00 3F	0000 dddd	Chorus Parameter 15 (12768 - 52768)					
			-20000 - 20000					
#	00 40	0000 aaaa						
	00 41	0000 bbbb						
	00 42	0000 cccc						
	00 43	0000 dddd	Chorus Parameter 16 (12768 - 52768)					
			-20000 - 20000					
#	00 44	0000 aaaa						
	00 45	0000 bbbb						
	00 46	0000 cccc						
	00 47	0000 dddd	Chorus Parameter 17 (12768 - 52768)					
			-20000 - 20000					
#	00 48	0000 aaaa						
	00 49	0000 bbbb						
	00 4A	0000 cccc						
	00 4B	0000 dddd	Chorus Parameter 18 (12768 - 52768)					
			-20000 - 20000					

* appendix 1		
Category	Type	Value
Thru	Thru	0
Filter	Equalizer	1
	Mid-Side EQ	77
	Spectrum	2
	Isolator	3
	Low Boost	4
	SuperFilter	5
	MM Filter	88
	Step Filter	6
	Enhancer	7
	Exciter	92
	Auto Wah	8
	Humanizer	9
Phaser	Phaser	11
	SmallPhaser	12
	Script 90	13
	Script 100	90
	Step Phaser	14
	M StagePhsr	15
	Inf Phaser	16
Flanger	Flanger	24
	SBF-325	73
	StepFlanger	25
Chorus	Chorus	23
	Hexa-Chorus	26
	Trem Chorus	27
	Space-D	28
	CE-1	72
	SDD-320	74
	JUNO Chorus	87
Modulation	Ring Mod	17
	Tremolo	18
	Auto Pan	19
	Slicer	20
	Rotary	21
	VK Rotary	22
Drive / Amp	Overdrive	29
	Distortion	30
	T-Scream	31
	Fuzz	86
	Fattener	79
	HMS Distort	89
	Saturator	84
	W Saturator	85
	Gt Amp Sim	32
	EP Amp Sim	81
	Speaker Sim	10
Comp / Limiter	Compressor	33
	M/S Comp	78
	Limiter	34
	Sustainer	35
	Transient	76
	Gate	36
Delay	Delay	37
	Mod Delay	38
	2Tap PanDly	75
	3Tap PanDly	39
	4Tap PanDly	40
	MultiTapDly	41
	Reverse Dly	42
	TimeCtrlDly	43
	Tape Echo	44
	M/S Delay	80
Looper	DJFX Looper	82
	BFM Looper	83
Lo-fi	LOFI Comp	45
	Bit Crusher	46
	Phonograph	91
Pitch	PitchShiftr	47
	2V PShifter	48
Combination	OD->Chorus	49
	OD->Flanger	50
	OD->Delay	51
	DS->Chorus	52
	DS->Flanger	53
	DS->Delay	54
	OD/DS>T.Wah	55
	OD/DS>A.Wah	56
	Gt->Chorus	57
	Gt->Flanger	58
	Gt->Phaser	59

	Gt->Delay		60	
	EP->Tremolo		61	
	EP->Chorus		62	
	EP->Flanger		63	
	EP->Phaser		64	
	EP->Delay		65	
	Enhncr->Cho		66	
	Enhncr->Fl		67	
	Enhncr->Dly		68	
	Chorus->Dly		69	
	Flanger->Dly		70	
	Chorus->Fl		71	
	JD-Multi		93	

4. Supplementary Material

■ Decimal and Hexadecimal Table

(An "H" is appended to the end of numbers in hexadecimal notation.)

In MIDI documentation, data values and addresses/sizes of Exclusive messages, etc. are expressed as hexadecimal values for each 7 bits. The following table shows how these correspond to decimal numbers.

D	H	D	H	D	H	D	H
0	00H	32	20H	64	40H	96	60H
1	01H	33	21H	65	41H	97	61H
2	02H	34	22H	66	42H	98	62H
3	03H	35	23H	67	43H	99	63H
4	04H	36	24H	68	44H	100	64H
5	05H	37	25H	69	45H	101	65H
6	06H	38	26H	70	46H	102	66H
7	07H	39	27H	71	47H	103	67H
8	08H	40	28H	72	48H	104	68H
9	09H	41	29H	73	49H	105	69H
10	0AH	42	2AH	74	4AH	106	6AH
11	0BH	43	2BH	75	4BH	107	6BH
12	0CH	44	2CH	76	4CH	108	6CH
13	0DH	45	2DH	77	4DH	109	6DH
14	0EH	46	2EH	78	4EH	110	6EH
15	0FH	47	2FH	79	4FH	111	6FH
16	10H	48	30H	80	50H	112	70H
17	11H	49	31H	81	51H	113	71H
18	12H	50	32H	82	52H	114	72H
19	13H	51	33H	83	53H	115	73H
20	14H	52	34H	84	54H	116	74H
21	15H	53	35H	85	55H	117	75H
22	16H	54	36H	86	56H	118	76H
23	17H	55	37H	87	57H	119	77H
24	18H	56	38H	88	58H	120	78H
25	19H	57	39H	89	59H	121	79H
26	1AH	58	3AH	90	5AH	122	7AH
27	1BH	59	3BH	91	5BH	123	7BH
28	1CH	60	3CH	92	5CH	124	7CH
29	1DH	61	3DH	93	5DH	125	7DH
30	1EH	62	3EH	94	5EH	126	7EH
31	1FH	63	3FH	95	5FH	127	7FH

D: decimal

H: hexadecimal

* Decimal values such as MIDI channel, bank select, and program change are listed as one greater than the values given in the above table.

* A 7-bit byte can express data in the range of 128 steps. For example, two hexadecimal numbers aa bbH expressing two 7-bit bytes would indicate a value of

aa x 128+bb.

* In the case of values which have a +/- sign, 00H = -64, 40H = +/-0, and 7FH = +63, so that the decimal expression would be 64 less than the value

given in the above chart. In the case of two types, 00 00H = -8192, 40 00H = +/-0, and 7F 7FH = +8191.

For example, if aa bbH were expressed as decimal, this would be aa bbH - 40 00H = aa x 128+bb - 64 x 128.

* Data marked "Use nibbled data" is expressed in hexadecimal in 4-bit units. A value expressed as a 2-byte nibble 0a 0bH has the value of a x 16+b.

<Example1> What is the decimal expression of 5AH?

From the preceding table, 5AH = 90

<Example2>

What is the decimal expression of the value 12 34H given as hexadecimal for each 7 bits?

From the preceding table, since 12H = 18 and 34H = 52

18 x 128+52 = 2356

<Example3>

What is the decimal expression of the nibbled value 0A 03 09 0D?

From the preceding table, since 0AH = 10, 03H = 3, 09H = 9, 0DH = 13

((10 x 16+3) x 16+9) x 16+13 = 41885

<Example4> What is the nibbled expression of the decimal value 1258?

16) 1258

16) 78 ...10

16) 4 ...14

0 ... 4

Since from the preceding table, 0 = 00H, 4 = 04H, 14 = 0EH, 10 = 0AH, the result is:

00 04 0E 0AH

■ Examples of Actual MIDI Messages

<Example1> 92 3E 5F

9n is the Note-on status, and n is the MIDI channel number. Since 2H = 2, 3EH = 62, and 5FH = 95, this is a Note-on message with MIDI CH = 3, note number 62 (note name is D4), and velocity 95.

<Example2> CE 49

CnH is the Program Change status, and n is the MIDI channel number. Since EH = 14 and 49H = 73, this is a Program Change message with MIDI CH = 15, program number 74.

<Example3> EA 00 28

EnH is the Pitch Bend Change status, and n is the MIDI channel number. The 2nd byte (00H = 0) is the LSB and the 3rd byte (28H = 40) is the MSB, but Pitch Bend Value is a signed number in which 40 00H (= 64 x 12+80 = 8192) is 0, so this Pitch Bend Value is

28 00H - 40 00H = 40 x 12+80 - (64 x 12+80) = 5120 - 8192 = -3072

If the Pitch Bend Sensitivity is set to 2 semitones, -8192 (00 00H) will cause the pitch to change -200 cents, so in this case -200 x (-3072) / (-8192) = -75 cents of Pitch Bend is being applied to MIDI channel 11.

<Example4> B3 64 00 65 00 06 0C 26 00 64 7F 65 7F

BnH is the Control Change status, and n is the MIDI channel number. For Control Changes, the 2nd byte is the control number, and the 3rd byte is the value.

In a case in which two or more messages consecutive messages have the same status, MIDI has a provision called "running status" which allows the status

byte of the second and following messages to be omitted. Thus, the above messages have the following meaning.

```
B3 64 00 MIDI ch.4, lower byte of RPN parameter number: 00H
(B3) 65 00 (MIDI ch.4) upper byte of RPN parameter number: 00H
(B3) 06 0C (MIDI ch.4) upper byte of parameter value: 0CH
(B3) 26 00 (MIDI ch.4) lower byte of parameter value: 00H
(B3) 64 7F (MIDI ch.4) lower byte of RPN parameter number: 7FH
(B3) 65 7F (MIDI ch.4) upper byte of RPN parameter number: 7FH
```

In other words, the above messages specify a value of 0C 00H for RPN parameter number 00 00H on MIDI channel 4, and then set the RPN parameter number to 7F 7FH.

RPN parameter number 00 00H is Pitch Bend Sensitivity, and the MSB of the value indicates semitone units, so a value of 0CH = 12 sets the maximum pitch bend range to +/-12 semitones (1 octave). (On GS sound generators the LSB of Pitch Bend Sensitivity is ignored, but the LSB should be transmitted anyway (with a value of 0) so that operation will be correct on any device.)

Once the parameter number has been specified for RPN or NRPN, all Data Entry messages transmitted on that same channel will be valid, so after the desired value has been transmitted, it is a good idea to set the parameter number to 7F 7FH to prevent accidents. This is the reason for the (B3) 64 7F (B3) 65 7F at the end.

It is not desirable for performance data (such as Standard MIDI File data) to contain many events with running status as given in <Example 4>. This is because if playback is halted during the song and then rewound or fastforwarded, the sequencer may not be able to transmit the correct status, and the sound generator will then misinterpret the data. Take care to give each event its own status.

It is also necessary that the RPN or NRPN parameter number setting and the value setting be done in the proper order. On some sequencers, events occurring in the same (or consecutive) clock may be transmitted in an order different than the order in which they were received. For this reason it is a good idea to slightly skew the time of each event (about 1 tick for TPQN = 96, and about 5 ticks for TPQN = 480).

* TPQN: Ticks Per Quarter Note

■ Example of an Exclusive Message and Calculating a Checksum

Roland Exclusive messages (RQ1, DT1) are transmitted with a checksum at the end (before F7) to make sure that the message was correctly received. The value of the checksum is determined by the address and data (or size) of the transmitted Exclusive message.

● How to calculate the checksum

(hexadecimal numbers are indicated by "H")

The checksum is a value derived by adding the address, size, and checksum itself and inverting the lower 7 bits.

Here's an example of how the checksum is calculated. We will assume that in the Exclusive message we are transmitting, the address is aabbccddH and the data or size is eeffH.

```
aa + bb + cc + dd + ee + ff = sum
sum ? 128 = quotient ... remainder
128 - remainder = checksum
```

<Example> Setting program level to 74 (DT1).

According to the Parameter Address Map, the start address of Temporary Program is 01 00 00 00H, the offset address of Program common is 00 00 00H, and the address of Program level is 00 10H. Therefore the address is;

```
01 00 00 00H
00 00 00H
+) 00 10H
```

```
-----
01 00 00 10H
```

Level 74 of the Program has the value of 4AH. So the system exclusive message should be sent is;

```
F0 41 10 00 00 00 12 12 01 00 00 10 4A ?? F7
(1) (2) (3) (4) (5) address data checksum (6)
```

(1) Exclusive Status (2) ID (Roland) (3) Device ID (17)
(4) Model ID (JUNO-X) (5) Command ID (DT1) (6) End of Exclusive

Then calculate the checksum.

01H + 00H + 00H + 10H + 4AH = 1 + 0 + 0 + 16 + 74 = 91 (sum)

91 (sum) / 128 = 0 (quotient) ... 91 (remainder)

checksum = 128 ... 91 (remainder) = 37 = 25H

This means that F0 41 10 00 00 00 12 12 01 00 00 10 4A 25 F7 is the message should be sent.

ASCII Code Table

Program Name, etc., of MIDI data are described the ASCII code in the table below.

D	H	Char	D	H	Char	D	H	Char
32	20H	SP	64	40H	@	96	60H	'
33	21H	!	65	41H	A	97	61H	a
34	22H	"	66	42H	B	98	62H	b
35	23H	#	67	43H	C	99	63H	c
36	24H	\$	68	44H	D	100	64H	d
37	25H	%	69	45H	E	101	65H	e
38	26H	&	70	46H	F	102	66H	f
39	27H	'	71	47H	G	103	67H	g
40	28H	(	72	48H	H	104	68H	h
41	29H	)	73	49H	I	105	69H	i
42	2AH	*	74	4AH	J	106	6AH	j
43	2BH	+	75	4BH	K	107	6BH	k
44	2CH	,	76	4CH	L	108	6CH	l
45	2DH	-	77	4DH	M	109	6DH	m
46	2EH	.	78	4EH	N	110	6EH	n
47	2FH	/	79	4FH	O	111	6FH	o
48	30H	0	80	50H	P	112	70H	p
49	31H	1	81	51H	Q	113	71H	q
50	32H	2	82	52H	R	114	72H	r
51	33H	3	83	53H	S	115	73H	s
52	34H	4	84	54H	T	116	74H	t
53	35H	5	85	55H	U	117	75H	u
54	36H	6	86	56H	V	118	76H	v
55	37H	7	87	57H	W	119	77H	w
56	38H	8	88	58H	X	120	78H	x
57	39H	9	89	59H	Y	121	79H	y
58	3AH	:	90	5AH	Z	122	7AH	z
59	3BH	;	91	5BH	[	123	7BH	{
60	3CH	<	92	5CH	\	124	7CH	
61	3DH	=	93	5DH	]	125	7DH	}
62	3EH	>	94	5EH	^			
63	3FH	?	95	5FH	_			

D: decimal

H: hexadecimal

* "SP" is space.

5.MIDI Implementation Chart

[Synthesizer]
Model JUNO-X

MIDI Implementation Chart

Date : April 26, 2022
Version : 1.10

Function...		Transmitted	Recognized	Remarks
Basic	Default	1-16	1-16	
Channel	Changed	1-16	1-16	Memorized
Mode	Default	Mode 3	Mode 3	
	Messages	Mono, Poly	Mode 3, 4 (M=1)	*2
	Altered	*****		
Note		0-127	0-127	
Number:	True Voice	*****	0-127	
Velocity	Note On	o	o	
	Note Off	o	o	
After	Key's	x	o	*1
Touch	Channel's	o	o	*1
Pitch Bend		o	o	*1
Control	0,32	o	o	*1 Bank select
Change	1	o	o	*1 Modulation
	2	o	o	Breath type
	4	o	o	Foot type
	5	o	o	Portamento time
	6,38	o	o	Data entry
	7	o	o	*1 Volume
	10	o	o	*1 Panpot
	11	o	o	*1 Expression
	16	o	o	General purpose controller 1
	17	o	o	General purpose controller 2
	18	o	o	General purpose controller 3
	19	o	o	General purpose controller 4
	64	o	o	*1 Hold 1
	65	o	o	Portamento
	66	o	o	Sostenuto
	67	o	o	Soft
	68	o	o	Legato foot switch
	71	o	o	Resonance
	72	o	o	Release time
	73	o	o	Attack time
	74	o	o	Cutoff
	75	o	o	Decay time
	76	o	o	Vibrate rate
	77	o	o	Vibrate depth
	78	o	o	Vibrate delay
	80	o	o	General purpose controller 5
	81	o	o	General purpose controller 6
	82	o	o	General purpose controller 7
	83	o	o	General purpose controller 8
	84	o	o	Portamento control
	91	o	o (Reverb)	General purpose effects 1
	93	o	o (Chorus)	General purpose effects 3
	1-31,33-95	o	-	General purpose controller
	98,99	x	x	NRPN LSB,MSB
	100,101	x	o	RPN LSB,MSB
Program		o	*1 o	*1
Change	:True Number	*****	0-127	Program No.1-128
System Exclusive		o	*3 o	*1
System	:Song Position	x	x	
Common	:Song Select	x	x	
	:Tune Request	x	x	
System	:Clock	o	o	
Real Time	:Commands	x	x	
	:All Sound Off	x	o	
	:Reset All Controllers	x	o	
Aux	:Local On/Off	x	x	
Messages	:All Notes Off	x	o (123-127)	
	:Active Sensing	o	o	
	:System Reset	x	x	
Notes		*1 o x is selectable. *2 Recognized as M=1 even if M#1. *3 Transmitted only when "Transmitted Edit Data" is ON or RQ1 is received.		
Mode 1 : OMNI ON, POLY	Mode 2 : OMNI ON, MONO	o : Yes		
Mode 3 : OMNI OFF, POLY	Mode 4 : OMNI OFF, MONO	x : No		